

Algebraic Topology 1

Labix

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1 Introduction to Homotopy Theory and the Fundamental Group

1.1 Homotopy between two Maps

In general, showing that two spaces are not homeomorphic is hard. Therefore we turn to a weaker statement: homotopy equivalence. It will become apparent that homeomorphic spaces are necessarily homotopic, but homotopic spaces may not be homeomorphic in general.

Definition 1.1.1 (Homotopy)

Let X, Y be spaces. Let $f, g : X \rightarrow Y$ be maps. A homotopy from f to g is a continuous function $H : X \times I \rightarrow Y$ such that

$$H(x, 0) = f(x) \quad \text{and} \quad H(x, 1) = g(x)$$

for all $x \in X$. In this case, we say that f is homotopic to g and write $f \simeq g$.

In particular, we require H to be continuous in the sense that both on any point $x \in X$ and any $t \in I$ so that the transformation taken between two functions are continuous.

Proposition 1.1.2

Let X, Y be spaces. Then the relation of homotopy on $\{f : X \rightarrow Y \mid f \text{ is continuous}\}$ is an equivalence relation.

Proof

Let $f : X \rightarrow Y$ be a continuous function. By defining $H(x, t) = f(x)$, f is homotopic to itself. Now if $f, g : X \rightarrow Y$ are homotopic maps, this means that there is a homotopy $H : X \times I \rightarrow Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. Now consider the map $H' : X \times I \rightarrow Y$ by

$$H'(x, t) = H(x, 1 - t)$$

Then $H'(x, 0) = g(x)$ and $H'(x, 1) = f(x)$. Moreover since $t \rightarrow 1 - t$ is continuous and the composition of continuous functions is continuous, H' is continuous and we are done.

Finally suppose that $f, g : X \rightarrow Y$ are homotopic with homotopy H_1 and $g, h : X \rightarrow Y$ are homotopic with homotopy H_2 . Then define

$$H_3(x, t) = \begin{cases} H_1(x, 2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ H_2(x, 2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

Then this map is continuous by the pasting lemma. ■

Definition 1.1.3 (The Set of Homotopy Classes of Maps)

Let X, Y be spaces. Define the set of homotopy classes of maps from X to Y to be

$$[X, Y] = \frac{\{f : X \rightarrow Y \mid f \text{ is continuous}\}}{\sim}$$

where $f \sim g$ if f and g are homotopic.

The concept of a homotopy allows us to relax the condition on homeomorphism.

Definition 1.1.4 (Homotopy Equivalence)

Let X and Y be spaces. We say that X and Y are homotopy equivalent if there exists maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $g \circ f \simeq I_X$ and $f \circ g \simeq I_Y$. In this case g is said to be the homotopic inverse of f .

We often simply say that X and Y are homotopic instead of homotopy equivalent in this case.

Homeomorphic spaces are necessarily homotopy equivalent. Indeed If f is a homeomorphism with g as an inverse, then $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$ so that homeomorphism is a stronger property than

homotopic. On the contrary, homotopic spaces are not necessarily homeomorphic. However, as stated before, showing that two spaces are not homeomorphic is a hard problem.

Lemma 1.1.5 Let $f_{1,2} : X \rightarrow Y$ be homotopic maps. Let $g_{1,2} : Y \rightarrow Z$ be homotopic maps. Then $g_1 \circ f_1, g_2 \circ f_2 : X \rightarrow Z$ is homotopic.

Proof Suppose that $F : X \times I \rightarrow Y$ gives the homotopy for $f_{1,2}$ and $G : Y \times I \rightarrow Z$ gives the homotopy for $g_{1,2}$. Define

$$H(x, t) = G(F(x, t), t)$$

This is a composition of continuous functions and thus is continuous. Moreover, we have that $H(x, 0) = G(F(x, 0), 0) = g_1(f_1(x))$ and similarly $H(x, 1) = g_2(f_2(x))$ thus we are done. ■

Definition 1.1.6 (Null-Homotopic Maps) Let $f : X \rightarrow Y$ be a map of spaces. We say that f is null-homotopic if it is homotopic to a constant map.

Retractions and deformation retracts are closely related to homotopy. In particular, deformation retracts provide us with a large amount of examples of homotopically equivalent spaces. We first define the notion of a retract.

Definition 1.1.7 (Retract) We say that $A \subset X$ is a retract of X a topological space if there is a continuous function $r : X \rightarrow A$ such that $r(a) = a$ for all $a \in A$.

In particular, notice that $r \circ \iota_A : A \rightarrow A$ is the identity map on A .

Definition 1.1.8 (Deformation Retracts) Let X be a topological space and $A \subset X$. We say that a homotopy $H : X \times I \rightarrow X$ is a deformation retract of X to A if the following are true.

- $H(x, 0) = \text{id}_X$ is the identity function on X
- $H(x, 1) \in A$ for all $x \in X$
- $H(a, 1) = a$ for all $a \in A$

In other words, it is a homotopy from the identity of X to a retract of X to A .

The difference between retracts and deformation retracts is that deformation retracts is time dependent. This allows us to track where points are mapped to through time. Compared to retracts, we can only see where each point is sent, instead of the path they trace out as they retract.

This idea is made concrete by the following lemma.

Lemma 1.1.9 Let $H : X \times I \rightarrow X$ be a deformation retract. Then $\iota_A \circ r \simeq \text{id}_X$.

Proof Clearly if H is a deformation retract that $\iota_A \circ r \simeq \text{id}_X$ by definition. ■

Deformation retracts are thus stronger than retracts in the following sense: For retracts we can only deduce that $r \circ \iota_A = \text{id}_A$. For deformation retracts, since $r = H(-, 1) : X \rightarrow A$ is a retract, not only do we have

$$H(-, 1) \circ \iota_A = r \circ \iota_A = \text{id}_A$$

we also have that $\iota_A \circ r \simeq \text{id}_X$. We shall use this fact to show that deformation retracts induces isomorphisms in fundamental groups. This is also in part due to the following lemma.

Lemma 1.1.10 If $H : X \times I \rightarrow X$ is a deformation retract from X to $A \subset X$, then X and A are homotopy equivalent.

Proof H being a deformation retract implies that $\iota_A \circ r \simeq \text{id}_X$. We just need to construct $F : A \times I \rightarrow A$ such that $r \circ \iota_A \simeq \text{id}_A$. But clearly $r \circ \iota_A$ is the identity map on A . This means trivially that $\text{id}_A \simeq \text{id}_A$ thus we are done. ■

Definition 1.1.11 (Contratible Space) A space X is contractible if it is homotopy equivalent to a point.

It is easy to see that X is contractible if the identity map on X is null-homotopic.

Definition 1.1.12 (Locally Contractible Space) A topological space X is locally contractible if for every $x \in X$, there exist an open neighbourhood U of x such that there exists an open set $V \subset U$ containing x such that V is contractible.

This is a good place to mention that whenever there is a local version of a topological property, it most likely means that every point in the space has a neighbourhood for which an open subset of the neighbourhood has the property. In our case, contractible would be the topological property.

It is very untrue that the local topological property on every point implies the topological property globally, nor is it true that the global property implies the local version.

1.2 The Path Connected Components

Definition 1.2.1 (The Path Connected Components) Let X be a space. Define the path connected components of X to be

$$\pi_0(X) = \{\text{Path components of } X\}$$

Proposition 1.2.2 Let X be a space. Define a relation on X where $x \sim y$ if and only if there is a path from x to y . Then $\pi_0(X) = \frac{X}{\sim}$.

1.3 Homotopies Relative to a Subset

Definition 1.3.1 (Homotopy Relative to a Subset) Let X, Y be a space and $A \subseteq X$ a subset of X . We say that a homotopy $H : X \times I \rightarrow Y$ is a homotopy relative to A if

$$H(a, t) = f(a) = g(a)$$

for all $a \in A$ and all $t \in I$. This means that the homotopy fixes all the values in A . In this case, we write $f \stackrel{A}{\simeq} g$.

Proposition 1.3.2

$\stackrel{A}{\simeq}$ is a homotopy equivalence.

1.4 Pointed Spaces and Pointed Homotopies

Definition 1.4.1 (Pointed Spaces)

A pointed space is a pair (X, x_0) where X is a space and $x_0 \in X$.

Definition 1.4.2 (Pointed Maps)

Let (X, x_0) and (Y, y_0) be pointed spaces. Let $f : X \rightarrow Y$ be a map. We say that f is pointed if $f(x_0) = y_0$.

Definition 1.4.3 (Pointed Homotopies)

Let (X, x_0) and (Y, y_0) be pointed spaces. Let $f, g : X \rightarrow Y$ be pointed maps. A pointed homotopy from f to g is a homotopy $H : X \times I \rightarrow Y$ from f to g relative to $\{x_0\}$.

Definition 1.4.4 (The Set of Pointed Homotopy Classes of Maps)

Let $(X, x_0), (Y, y_0)$ be spaces. Define the set of homotopy classes of maps from (X, x_0) to (Y, y_0) to be

$$[(X, x_0), (Y, y_0)]_* = \frac{\{f : (X, x_0) \rightarrow (Y, y_0) \mid f \text{ is a pointed map}\}}{\sim}$$

where $f \sim g$ if there exists a pointed homotopy from f to g .

1.5 The Algebra of Paths**Definition 1.5.1** (The Constant Path)

Let X be a space. Let $x_0 \in X$. Define the constant path of X at x_0 to be the path $e_{x_0} : I \rightarrow X$ defined by

$$e_{x_0}(t) = x_0$$

Definition 1.5.2 (Orientation Reversal) Let X be a space. Let $f : I \rightarrow X$ be a path in X . Denote

$$\bar{f}(t) = f(1 - t)$$

for $t \in [0, 1]$ the reverse traversal of the path f .

Definition 1.5.3 (Concatenation of Paths) Let $f, g : [0, 1] \rightarrow X$ be continuous paths to a topological space X such that $f(1) = g(0)$. Then the concatenation of f and g can be defined:

$$(f \cdot g)(t) = \begin{cases} f(2t) & 0 \leq t \leq \frac{1}{2} \\ g(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

The product of paths is read left to right. This is contradictory to the composition of functions, in which the operation is done from right to left. Indeed, the concatenation of paths is not a composition of functions and one should be clear that they are different concepts.

Essentially, the concatenation of paths increases the velocity of the two paths in which we are travelling so that after traversing on f , we traverse on g and the resulting domain is kept at the unit interval I .

We can show that the concatenation of paths is well defined under the equivalence class, which is another important observation.

Lemma 1.5.4 Suppose that $f_1, f_2, g_1, g_2 : [0, 1] \rightarrow X$ with $f_1 \simeq f_2$ and $g_1 \simeq g_2$ and $f_1(1) = f_2(1) = g_1(0) = g_2(0)$. Then

$$f_1 \cdot g_1 \simeq f_2 \cdot g_2$$

Proof Suppose that $f_1 \simeq f_2$ with homotopy map F . Suppose that $g_1 \simeq g_2$ with homotopy map G . Define

$$H(s, t) = \begin{cases} F(2s, t) & \text{if } 0 \leq s \leq \frac{1}{2} \\ G(2s - 1, t) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

This map is continuous by the pasting lemma. Moreover, it satisfies the definition of a homotopy between $f_1 \cdot g_1$ and $f_2 \cdot g_2$ thus we are done. ■

Through this lemma, it now makes sense to talk about concatenation in equivalence classes of paths since it does not matter what representative we choose.

Proposition 1.5.5 Let $f : [0, 1] \rightarrow X$ be a path such that $f(0) = x$ and $f(1) = y$. Then

$$e_x \cdot f \simeq f \quad \text{and} \quad f \cdot e_y \simeq f$$

Proof Using the homotopy $H : I \times I \rightarrow X$ defined by

$$H(s, t) = \begin{cases} e_x(s) = x & \text{if } 0 \leq s \leq \frac{1-t}{2} \\ f\left(\frac{2s+t-1}{t+1}\right) & \text{if } \frac{1-t}{2} \leq s \leq 1 \end{cases}$$

we see that indeed, $H(s, 0) = (e_x \cdot f)(s)$ and $H(s, 1) = f(s)$. By the pasting lemma, H is continuous and we are done.

Similarly, we see that

$$H(s, t) = \begin{cases} f\left(\frac{2s}{t+1}\right) & \text{if } 0 \leq s \leq \frac{t+1}{2} \\ e_y(s) = y & \text{if } \frac{t+1}{2} \leq s \leq 1 \end{cases}$$

■

1.6 The Fundamental Group

Definition 1.6.1 (Loops of a Space)

Let X be a space. Let $f : I \rightarrow X$ be a path. We say that f is a loop if $f(0) = f(1)$.

Definition 1.6.2 (The Fundamental Group)

Let (X, x_0) be a pointed space. Define the fundamental group of (X, x_0) to be the set

$$\pi_1(X, x_0) = \frac{\{f : (I, 0) \rightarrow (X, x_0) \mid f \text{ is a pointed map such that } f \text{ is a loop}\}}{\sim}$$

where $f \sim g$ if there exists a pointed homotopy equivalence from f to g , together with the binary operation given by

$$[f] \cdot [g] = [f \cdot g]$$

Lemma 1.6.3

Let (X, x_0) be a pointed space. Then $\pi_1(X, x_0)$ is a group.

Lemma 1.6.4

Let (X, x_0) be a pointed space. Then there is a bijection of sets

$$\pi_1(X, x_0) = [(S^1, 1), (X, x_0)]_*$$

Recall that the notion of homotopy is to attempt to classify topological spaces up to homeomorphism. The big question is whether the fundamental group is invariant under homeomorphism. Before we answer this question, we first look at how the fundamental group reacts to a continuous map.

Theorem 1.6.5 (Induced Homomorphisms) Let $\phi : (X, x_0) \rightarrow (Y, y_0)$ be a continuous mapping between two topological spaces. Then ϕ induces a group homomorphism

$$\phi_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

defined by $\phi_*([f]) = [\phi \circ f]$ for $[f] \in \pi_1(X, x_0)$. Moreover, ϕ_* satisfies the following functorial properties:

- If $\psi : (Y, y_0) \rightarrow (Z, z_0)$ is a map then $(\psi \circ \phi)_* = \psi_* \circ \phi_*$
- $(\text{id}_{(X, x_0)})_* = \text{id}_{\pi_1(X, x_0)}$

Finally, if ϕ is a homeomorphism, then ϕ_* is a group isomorphism.

Proof We first show that ϕ_* is a well defined map. This means that if f, g are loops based in X with $f \simeq g$, then $\phi \circ f \simeq \phi \circ g$. If $H : I \times I \rightarrow X$ is a homotopy from f to g , then $G = \phi \circ H$ is a homotopy from $\phi \circ f$ to $\phi \circ g$ so this is clear. To show that it is a group homomorphism, notice

that

$$\phi_*([f] \cdot [g]) = \phi_*([f \cdot g]) = [\phi \circ (f \cdot g)]$$

and

$$\phi_*([f]) \cdot \phi_*([g]) = [\phi \circ f] \cdot [\phi \circ g]$$

So we just have to show that $\phi \circ (f \cdot g) \simeq (\phi \circ f) \cdot (\phi \circ g)$. But by definition,

$$(\phi \circ f) \cdot (\phi \circ g)(t) = \begin{cases} \phi \circ f(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ \phi \circ g(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

which is just the same as $\phi \circ (f \cdot g)$.

For the remaining functorial properties, we have that

- Clearly we have that for any $[f] \in \pi_1(X, x_0)$,

$$\begin{aligned} (\psi \circ \phi)_*([f]) &= [\psi \circ \phi \circ f] \\ &= \psi_*([\phi \circ f]) \\ &= \psi_*(\phi_*([f])) \\ &= (\psi_* \circ \phi_*)([f]) \end{aligned}$$

- Trivial

Finally, if ϕ is a homeomorphism, we have that $(\text{id}_{(X, x_0)})_* = (\phi^{-1})_* \circ \phi_*$ and similarly $(\text{id}_{(X, x_0)})_* = \phi_* \circ (\phi^{-1})_*$ which shows that ϕ_* has an inverse. ■

Notice that the above theorem implies that $\pi_1(\cdot)$ is a functor (for category theorists) from pointed topological spaces to groups. This functor is called the fundamental group. However the following theorem is of greater importance than the fact that homeomorphic maps have isomorphic fundamental groups.

Theorem 1.6.6 (Homotopy Invariance) Let $\phi, \psi : (X, x_0) \rightarrow (Y, y_0)$ be continuous maps. Suppose that ϕ_* and ψ_* are the induced homomorphisms respectively. Then the following are true.

- If ϕ and ψ are homotopic, then $\phi_* = \psi_*$
- If ϕ is a homotopy equivalence, then ϕ_* is a group isomorphism

Proof

- Since ϕ and ψ are homotopic, then there exists $H(x, t) : X \times I \rightarrow Y$ such that H is continuous, $H(x, 0) = \phi(x)$ and $H(x, 1) = \psi(x)$. It also satisfies $H(-, t) = y_0$. Our goal is to show that $\phi \circ f$ is path homotopic to $\psi \circ f$ for any $f \in \pi_1(X, x_0)$. Define $F : I \times I \rightarrow Y$ by $F(x, t) = H(f(x), t)$. Then F is continuous. Moreover we have that $F(x, 0) = H(f(x), 0) = \phi(f(x))$. Moreover we have that $F(x, 1) = H(f(x), 1) = \psi(f(x))$. We also have that $F(0, t) = H(f(0), t) = y_0 = H(f(1), t) = F(1, t)$ which means that F is a path homotopy.
- Suppose that ψ is the homotopy inverse of ϕ such that $\psi \circ \phi \simeq I_{(X, x_0)}$ and $\phi \circ \psi \simeq I_{(Y, y_0)}$. Then by the above, we have that $\psi_* \circ \phi_* = I_{\pi_1(X, x_0)}$ and $\phi_* \circ \psi_* = I_{\pi_1(Y, y_0)}$ which means that ϕ_* and ψ_* are inverses of each other.

And so we conclude. ■

This shows that the fundamental group is an invariance of homotopic equivalence. In particular, topological spaces that are homotopic equivalent but not homeomorphic cannot be distinguished by the fundamental group.

In particular, as deformation retracts provide us with homotopy equivalence spaces, we have the following proposition, which asserts that the isomorphism of fundamental groups in fact comes from both the

retraction and the inclusion, often which the latter is more useful.

Proposition 1.6.7 Let $r : X \rightarrow A$ be a retraction from X to $A \subset X$. Let $i : A \rightarrow X$ be the inclusion. Then the following are true,

- r_* is surjective and i_* is injective
- If r is a deformation retract, then r_* and i_* are isomorphisms

Proof Recall that a map of sets $g \circ f$ has the property that g is surjective and f is injective if $g \circ f = \text{id}$.

- Since $r \circ i = I_A$, we have that $r_* \circ i_* = I_{\pi_1(A, x_0)}$ which means that at least i_* is injective and r_* is surjective.
- Deformation retracts imply homotopy equivalence from X to A . But by the above theorem, homotopy equivalence induces isomorphism in the fundamental group. This means that $i_* \circ r_* = I_{\pi_1(X, x_0)}$ which means that r is injective and i is surjective.

■

The proposition is useful in the following sense. If $r : X \rightarrow A$ is a retraction that the inclusion i_* is injective and r_* is surjective. If there is no group homomorphism for which r_* can be surjective or i_* can be injective, then we have reached a contradiction so that there is no retraction from X to A .

Theorem 1.6.8 (Change of Base Point) Let X be a space and let $x_0, x_1 \in X$ be such that they are connected by a path $u : I \rightarrow X$. Then this path induces a group isomorphism

$$\beta_u : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$$

given by $\beta_u([f]) = [\bar{u} \cdot f \cdot u]$. In particular,

- If $u \stackrel{\partial}{\simeq} u'$ are homotopic then $\beta_u = \beta_{u'}$
- $\beta_{e_{x_0}} = \text{id}_{\pi_1(X, x_0)}$
- If $v : I \rightarrow X$ is a path from x_1 to x_2 then $\beta_{u \cdot v} = \beta_v \circ \beta_u$

Proof We first show that β_u is an isomorphism. Suppose that $[\bar{u} \cdot f \cdot u] = [e_{x_0}]$ for $[f] \in \pi_1(X, x_0)$. Then it is easy to see that

$$\begin{aligned} [u] \cdot [\bar{u} \cdot f \cdot u] \cdot [\bar{u}] &= [u] \cdot [e_{x_0}] \cdot [\bar{u}] \\ [f] &= [e_{x_0}] \end{aligned}$$

Thus β_u is injective. For every $[g] \in \pi_1(X, x_1)$, it is clear that $[u \cdot g \cdot \bar{u}] \in \pi_1(X, x_0)$ is such that $\beta_u([u \cdot g \cdot \bar{u}]) = [g]$. Thus β_u is surjective.

For the remaining functorial properties, we have that

- If $u \stackrel{\partial}{\simeq} u'$ then $[u] = [u']$. This means that for any $[f] \in \pi_1(X, x_0)$, we have

$$[\bar{u}] \cdot [f] \cdot [u] = [\bar{u'}] \cdot [f] \cdot [u']$$

thus $\beta_u([f]) = \beta_{u'}([f])$ for any $[f] \in \pi_1(X, x_0)$.

- Clearly, $\beta_{e_{x_0}}([f]) = [\bar{e_{x_0}} \cdot f \cdot e_{x_0}] = [f]$ so that $\beta_{e_{x_0}}$ is the identity.
- Let $[f] \in \pi_1(X, x_0)$. Then we have

$$\begin{aligned} \beta_{u \cdot v}([f]) &= [\bar{u \cdot v} \cdot f \cdot u \cdot v] \\ &= [\bar{v} \cdot \bar{u} \cdot f \cdot u \cdot v] \\ &= [\bar{v}] \cdot \beta_u([f]) \cdot [v] \\ &= \beta_v(\beta_u([f])) \end{aligned}$$

so that $\beta_{u \cdot v} = \beta_v \circ \beta_u$.

And so we are done. ■

This means that the base point does not really matter, as long as they are path connected. In particular, if the entirety of X is path connected, then they really only have one unique fundamental group. However, beware that in the case that the change of base point can be given by two non-homotpic maps. This means that two different paths could potentially give rise to two different isomorphisms. It is common in this case to say that there is no canonical isomorphism.

Proposition 1.6.9 Let (X, x_0) and (Y, y_0) be pointed topological spaces. Then

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

In particular, notice that elements of the form $([f], [e_{y_0}])$ and $([e_{x_0}], [g])$ commute. This can be established by exhibiting homotopies from $([f], [g])$ to

$$([f], [e_{y_0}]) \cdot ([e_{x_0}], [g]) \quad \text{and} \quad ([e_{x_0}], [g]) \cdot ([f], [e_{y_0}])$$

respectively.

1.7 Simply Connected Spaces

Simply connected, spaces, also called 1-connected space, is a name given to when the space has trivial fundamental group. As we introduce the higher homotopy groups, there will be higher notions of connectivity that follow.

Definition 1.7.1 (Simply Connected) A space X is simply connected if it is path connected and its fundamental group

$$\pi_1(X, x_0) = \{e_{x_0}\}$$

is trivial on any base point.

Proposition 1.7.2 Contractible spaces are simply connected.

Proof Suppose that X is contractible. Then X is homotopy equivalent to a point. Any loop in the one point space must be a constant loop. ■

Lemma 1.7.3 Let X be a simply connected space. Let $f, g : I \rightarrow X$ be two paths that start at $x \in X$ and end at $y \in X$. Then f and g are homotopic.

Proof We have that

$$\begin{aligned} f \cdot \bar{g} &\simeq e_x \\ f \cdot \bar{g} \cdot g &\simeq e_x \cdot g \\ f &\simeq g \end{aligned}$$

and so we conclude. ■

We also have the local version of simply connectedness.

Definition 1.7.4 (Locally Simply Connected) A space X is said to be locally simply connected if for all $x \in X$, there exists a neighbourhood U of x such that there is an open set V in U for which V is contractible.

2 Covering Space Theory

General topology is concerned with the problem of whether certain types of maps can be lifted. This means that whether a map defined for a base space X could be extended to a larger space that contains the base space. Covering spaces arise in homotopy theory since they lift homotopies. However as it turns out, the theory of covering spaces are greatly related to the fundamental group of spaces.

2.1 Point Set Topology of Covering Spaces

We begin by studying covering spaces in a traditional manner. We investigate topological properties that the covering space share with the base space. Roughly speaking, covering spaces provide for each open set in the base space n disjoint sheets of cover that each map homeomorphically to the open set in the base space.

Definition 2.1.1 (Covering Space) A covering space is a map $p : \tilde{X} \rightarrow X$ such that for every point $x \in X$ there exists an open neighbourhood $x \in U \subset X$ such that

$$p^{-1}(U) = \coprod_{i \in I} V_i$$

is a disjoint union of open sets and that $p|_{V_i} : V_i \rightarrow U$ is a homeomorphism for every $i \in I$.

Some authors give a different definition for covering space. They are indeed equivalent.

Proposition 2.1.2 Let X be a space and let $p : Y \rightarrow X$ be a map. Then p is a covering space if and only if there exists an open cover $\{U_i \mid i \in I\}$ of X such that for each U in the open cover,

$$p^{-1}(U) = \coprod_{i \in I} V_i$$

is a disjoint union of open sets and that $p|_{V_i} : V_i \rightarrow U$ is a homeomorphism for every $i \in I$.

Proof Suppose that p is a covering space. Then for each $x \in X$ there exists an open neighbourhood U_x of x such that the condition of a covering space is satisfied. Then the collection $\{U_x \mid x \in X\}$ is an open cover of X satisfying the required condition.

Conversely, if there exists an open cover $\{U_i \mid i \in I\}$ of X for which the above condition is satisfied. Then for any $x \in X$, there exists U_i in the cover such that the covering space condition is satisfied. ■

This is even true without the assumption that p is a covering space. One can verify this by noticing that the proof did not involve the definition of a covering space. This will come up often in the study of local data. For example, in Manifolds or Algebraic Geometry. In general, suppose that we say a space X has property Q if for every $x \in X$, there exists an open neighbourhood U of x such that some condition C is satisfied. Then X has property Q if and only if there exists a covering of X called $\{U_i \mid i \in I\}$ in which condition C is satisfied by each U_i .

In point set topology, we defined the notion of a topological space, and then discussed how subsets and Cartesian products in sets can also be given the canonical structure of a space. Such idea is often considered in topology. For instance, we see below that subspaces and products of covering spaces are also covering space.

Proposition 2.1.3 Let X be a space and $p : \tilde{X} \rightarrow X$ a covering space. Let $A \subseteq X$ be a subspace of X and $\tilde{A} = p^{-1}(A)$. Then $p|_{\tilde{A}} : \tilde{A} \rightarrow A$ is a covering space of A .

Proof Let $x \in A$. Then there exists $U \subseteq X$ such that $p^{-1}(U) = \coprod_{i \in I} V_i$ for $V_i \cong U$. Now $U \cap A \subseteq A$. I claim that $U \cap A$ gives the required neighbourhood for $x \in A$ so that $p|_{\tilde{A}} : \tilde{A} \rightarrow A$ is a covering map. We have that

$$p^{-1}(U \cap A) = p^{-1}(U) \cap p^{-1}(A) = \left(\coprod_{i \in I} V_i \right) \cap \tilde{A} = \coprod_{i \in I} (V_i \cap \tilde{A})$$

Moreover, since $p|_{V_i} : V_i \rightarrow U$ is a homeomorphism, there is a homeomorphism

$$p|_{V_i \cap p^{-1}(A)} : V_i \cap p^{-1}(A) \rightarrow p(V_i \cap p^{-1}(A))$$

Since $p|_{V_i \cap p^{-1}(A)}$ is a bijection, we have a homeomorphism

$$p(V_i \cap p^{-1}(A)) \cong p(V_i) \cap p(p^{-1}(A)) = U \cap A$$

Thus we conclude. ■

Such a proof is omitted because it is often just a boring check that one understands the definition of covering spaces. The following proposition is also such an example.

Proposition 2.1.4 Let X, Y be spaces and $p_1 : \tilde{X} \rightarrow X$ and $p_2 : \tilde{Y} \rightarrow Y$ be covering spaces. Then $p_1 \times p_2 : \tilde{X} \times \tilde{Y} \rightarrow X \times Y$ is a covering space of $X \times Y$.

Proof Let $(x, y) \in X \times Y$. Then there exists an open neighbourhood U of x and C of y such that

$$p_1^{-1}(U) = \coprod_{i \in I} V_i \quad \text{and} \quad p_2^{-1}(C) = \coprod_{j \in J} D_j$$

such that $(p_1)|_{V_i} : V_i \xrightarrow{\cong} U$ and $(p_2)|_{D_j} : D_j \xrightarrow{\cong} C$. Now

$$p_1^{-1} \times p_2^{-1}(U \times C) = p_1^{-1}(U) \times p_2^{-1}(C) = \coprod_{i \in I} V_i \times \coprod_{j \in J} D_j = \coprod_{i \in I, j \in J} (V_i \times D_j)$$

Since $V_i \cong U$ and $D_j \cong C$ via p_1 and p_2 respectively, we conclude that $V_i \times D_j \cong U \times C$ via the map $p_1|_{V_i} \times p_2|_{D_j}$. ■

Compactness and Hausdorff is also preserved provided that each fiber is finite.

Proposition 2.1.5 Let X be a space and $p : \tilde{X} \rightarrow X$ be a covering space such that $|p^{-1}(x)|$ is finite for all $x \in X$. Then $\tilde{X}$ is compact Hausdorff if and only if X is compact Hausdorff.

Covering spaces are an example of local homeomorphisms. While homeomorphisms preserve global properties such as compactness and connectedness, local homeomorphisms preserve local properties such as locally simply connected or locally contractible.

2.2 Lifts and Covering Spaces

As mentioned before, covering spaces are interesting in the theory of homotopy because they lift homotopies. In this section we will see what this means precisely. We start with the notion of a general lift.

Definition 2.2.1 (Lifts) Let $f : X \rightarrow Y$ and $g : Z \rightarrow Y$ be continuous maps of spaces. We say that a map $h : X \rightarrow Z$ lifts f if $f = g \circ h$.

$$\begin{array}{ccc}
 & & Z \\
 & \nearrow h & \downarrow g \\
 X & \xrightarrow{f} & Y
 \end{array}$$

In some sense, a lift of a map $f : X \rightarrow Y$ is a factorization of the given map through a particular choice of factor $g : Z \rightarrow Y$. In the special case of lifting through a covering space, there are some consequences that we will need as we go further down the notes.

Proposition 2.2.2 Let $p : \tilde{X} \rightarrow X$ be a covering and $\tilde{f}, \tilde{g} : Y \rightarrow \tilde{X}$ maps. Then

- Trivial lift: $\tilde{f}$ is a lift of $p \circ \tilde{f}$
- Homotopies descend: If $\tilde{f} \simeq \tilde{g}$ then $p \circ \tilde{f} \simeq p \circ \tilde{g}$
- Paths descend: If $u, v : I \rightarrow \tilde{X}$ are paths with $u(1) = v(0)$, then

$$p \circ (u \cdot v) = (p \circ u) \cdot (p \circ v)$$

Proof

- $\tilde{f}$ is a lift of $p \circ \tilde{f}$ by definition
- Suppose that $H(x, t)$ is a homotopy from $\tilde{f}$ to $\tilde{g}$. Then the function $p \circ H$ is continuous since p and H are continuous. $p \circ H(x, 0) = p \circ \tilde{f}$ and $p \circ H(x, 1) = p \circ \tilde{g}$. Thus $p \circ H$ is a homotopy between $p \circ \tilde{f}$ and $p \circ \tilde{g}$.
- We have, by definition that

$$p \circ (u \cdot v)(t) = \begin{cases} p \circ u(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ p \circ v(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

This is precisely the definition of $(p \circ u) \cdot (p \circ v)$. ■

Since the coverings of a covering spaces are disjoint, lifts of functions to covering spaces enjoy the property of being unique so long they agree on at least one point. This is called the unique lifting property.

Proposition 2.2.3 (Unique Lifting Property) Let X be a space and $p : \tilde{X} \rightarrow X$ a covering space. Let $f : Y \rightarrow X$ be a map such that Y is connected. Let $\tilde{f}_1$ and $\tilde{f}_2$ both be lifts of f to $\tilde{X}$. Then the set

$$\{y \in Y \mid \tilde{f}_1(y) = \tilde{f}_2(y)\}$$

is either empty or is all of Y .

Proof It suffices to show that the set is both open and closed. Let $y \in Y$ and U be an open neighbourhood of $f(y) \in X$ for which $p^{-1}(U) = \coprod_{\alpha \in I} \tilde{U}_\alpha$. Suppose that $\tilde{f}_1$ is contained in $\tilde{U}_1$ and $\tilde{f}_2$ is contained in $\tilde{U}_2$.

If $\tilde{f}_1(y) = \tilde{f}_2(y)$ then $\tilde{U}_1 = \tilde{U}_2$. Thus on $N = \tilde{f}_1^{-1}(\tilde{U}_1) \cap \tilde{f}_2^{-1}(\tilde{U}_2)$, we have that

$$p|_{\tilde{U}_1} \circ \tilde{f}_1|_N = f|_N = p|_{\tilde{U}_2} \circ \tilde{f}_2|_N$$

But $p|_{\tilde{U}_1}$ is a homeomorphism. Thus $\tilde{f}_1|_N = \tilde{f}_2|_N$ and $y \in N \subset S$ so that S is open.

It remains to show that the set S is closed. Suppose for a contradiction that $y \in \bar{S}$ but $y \notin S$. This

means that $\tilde{f}_1(y) \neq \tilde{f}_2(y)$. Then it is clear that $\tilde{U}_1 \neq \tilde{U}_2$. Then

$$N = \tilde{f}_0^{-1}(\tilde{U}_1) \cap \tilde{f}_2^{-1}(\tilde{U}_2)$$

is an open neighbourhood of y . By definition of closure, $N \cap \bar{S}$ is non-empty. But this means that $\tilde{f}_1(N)$ and $\tilde{f}_2(N)$ must intersect, which is a contradiction of the definition of a covering space. ■

2.3 Lifting Properties and its Consequences

We now define formally what it means to lift a homotopy, and in particular paths for a general space.

Definition 2.3.1 (The Homotopy Lifting Property) Let $p : Z \rightarrow X$ be a map. We say that p has the homotopy lifting property (HLP) if for every homotopy $f_t : Y \rightarrow X$ and a lift $\tilde{f}_0 : Y \rightarrow Z$ of $f_0 : Y \rightarrow X$, there exists a unique homotopy $\tilde{f}_t : Y \rightarrow Z$ of $\tilde{f}_0$ that lifts f_t . In terms of diagrams, we have

$$\begin{array}{ccc} Y \times \{0\} & \xrightarrow{\tilde{f}_0} & Z \\ \downarrow i & \nearrow \tilde{f}_t & \downarrow p \\ Y \times I & \xrightarrow{f_t} & X \end{array}$$

This means that if f and g are homotopic functions from Y to X , then the lifts of f and g are also homotopic, given that they have the homotopy lifting property.

Definition 2.3.2 (The Path Lifting Property) Let $p : Z \rightarrow X$ be a map. We say that p has the path lifting property (PLP) if for any path $f : I \rightarrow X$ with $f(0) = x_0$ and $\tilde{x}_0 \in p^{-1}(x_0)$, there exists a unique lift $\tilde{f} : I \rightarrow Z$ of f such that $\tilde{f}_0 = \tilde{x}_0$. In terms of diagrams, we have

$$\begin{array}{ccc} & & Z \\ & \nearrow \tilde{f} & \downarrow p \\ I & \xrightarrow{f} & X \end{array}$$

Note that the path lifting property is just a special case of the homotopy lifting property since the homotopy could involve only a path instead of an entire space. In particular, choose Y as the one point space in the homotopy lifting property.

An important property of covering spaces is that they must have the homotopy lifting property. We need a lemma to prove it.

Lemma 2.3.3 Let $p : \tilde{X} \rightarrow X$ be a covering and let $H : Y \times I \rightarrow X$ be a homotopy. Let $g : Y \times \{0\} \rightarrow \tilde{X}$ be such that $p \circ g = H(y, 0)$. Then for every $y_0 \in Y$, there exists an open set N with $y_0 \in N \subset Y$ and a unique homotopy

$$\tilde{H}_N : N \times I \rightarrow \tilde{X}$$

such that $p \circ \tilde{H}_N = H|_{N \times I}$ and $\tilde{H}_N(y, 0) = g|_{N \times \{0\}}$. Moreover, if M is another neighbourhood, with $y_0 \in M \subset Y$, then

$$\tilde{H}_M|_{M \cap N \times I} = \tilde{H}_N|_{M \cap N \times I} = \tilde{H}_{M \cap N}$$

In other words, each $\tilde{H}_N(y, t)$ are lifts of $H(y, t)$ for fixed t with initial data g . Moreover, if N and M both subsets of Y then the two lifts $\tilde{H}_M$ and $\tilde{H}_N$ agree on their intersection.

Proof Let $p : \tilde{X} \rightarrow X$ be a covering map. Assume we have a homotopy $H : Y \times I \rightarrow X$ and $g : Y \times \{0\} \rightarrow \tilde{X}$ so that $p \circ g = H(y, 0)$. Let $\{U_\alpha\}$ be an open cover of X . By definition of a covering map, there exists a collection of disjoint subsets $\{V_\alpha^\beta\}$ of $\tilde{X}$ such that $p^{-1}(U_\alpha) = \bigsqcup_\beta V_\alpha^\beta$, and that

the restriction $p|_{V_\alpha^\beta} : V_\alpha^\beta \rightarrow U_\alpha$ is a homeomorphism. Denote $q_{\alpha,\beta} : U_\alpha \rightarrow V_\alpha^\beta$ the inverse of this homeomorphism.

Since H is continuous, for every $(y, t) \in Y \times I$ there exists $N \times (a, b) \subset Y \times I$ such that $F(y, t) \in U_\alpha$ for all $(y, t) \in N \times (a, b)$. Fixing y and using the fact that I is compact, we have finitely many $N_i \times (a_i, b_i)$ covering $\{y\} \times I$. Set $N = \bigcap_i N_i$. By property of a Lebesgue cover, we can choose $0 = t_0 < t_1 < \dots < t_n = 1$ such that $(t_i, t_{i+1}) \subset (a_i, b_i)$. Thus for every i , there is an α such that

$$F(N \times (t_i, t_{i+1})) \subset U_\alpha$$

I claim that there is a sequence of maps $\tilde{H}_N^k : N \times [0, t_k] \rightarrow \tilde{X}$ such that

- $\tilde{H}_N^k$ is a lift of $H|_{N \times [0, t_k]}$ (Indeed a lift)
- $(\tilde{H}_N^k)(y, 0) = g|_N$ (Agrees with Initial data)
- $\tilde{H}_N^{k+1}|_{N \times [0, t_k]} = \tilde{H}_N^k$ (Extension is natural)
- The above determines $\{\tilde{H}_N^k | 0 \leq k \leq n\}$ uniquely. (Uniqueness)

We construct the sequence by induction. When $k = 0$, we can clearly just set $\tilde{H}_N^0$ to be $g|_N$. Assume now that we have a sequence of maps $\tilde{H}_N^0, \dots, \tilde{H}_N^k$ by induction. By assumption, there exists some α such that $H(N \times [t_k, t_{k+1}]) \subset U_\alpha$. By allowing N to be smaller, we can assume that $\tilde{H}_N^k|_{N \times \{t_k\}}$ lies entirely in V_α^β for some β . Define

$$\tilde{F} = q_{\alpha,\beta} \circ H|_{N \times [t_k, t_{k+1}]}$$

Then clearly $\tilde{F}|_{N \times \{t_k\}} = \tilde{H}_N^k|_{N \times \{t_k\}}$. Now define

$$\tilde{H}_N^{k+1}(y, t) = \begin{cases} \tilde{H}_N^k(y, t) & \text{if } 0 \leq t \leq t_k \\ \tilde{F}(y, t) & \text{if } t_k \leq t \leq t_{k+1} \end{cases}$$

By the pasting lemma, $\tilde{H}_N^{k+1}$ is continuous. Then $\tilde{H}_N^n$ will have all the required properties and we are done.

To prove that two neighbourhoods has compatible lifts, suppose that we have two maps $\tilde{H}_N$ and $\tilde{H}'_N$ constructed. It is sufficient to show that for any $z \in N$, $\tilde{H}_N|_{\{z\} \times I} = \tilde{H}'_N|_{\{z\} \times I}$. As before, let $0 = t_0 < t_1 < \dots < t_m = 1$ be a partition such that $H(\{z\} \times [t_j, t_{j+1}]) \subset U_\alpha$. We proceed by induction. It is clear that both maps have to coincide on $\{z\} \times \{0\}$ as both have to match $g(z, 0)$ there. Assume that $\tilde{H}_N = \tilde{H}'_N$ on $[0, t_k]$. Since $[t_k, t_{k+1}]$ is connected, there exists a unique β such that $\tilde{H}_N(\{z\} \times [t_k, t_{k+1}])$ is contained in V_α^β . Similar there is a unique β' such that $\tilde{H}'_N(\{z\} \times [t_k, t_{k+1}]) \subset V_\alpha^{\beta'}$. But since they are assumed to be agree on (z, t_k) , we must have $\beta = \beta'$ and thus by construction of $\tilde{F}$, we must have that they coincide on $\{z\} \times [0, t_{k+1}]$. ■

With the above lemma, we can now prove that covering maps can lift homotopies.

Theorem 2.3.4 If $p : \tilde{X} \rightarrow X$ is a covering of X , then p has the homotopy lifting property and the path lifting property.

Proof Cover $Y \times I$ with open sets $N \times I$ as given by the above lemma. We then get a family of lifts $\tilde{H}_N : N \times I \rightarrow \tilde{X}$ that coincide on the intersections. By the pasting lemma, they are continuous and therefore lift $\tilde{H}$. ■

As a consequence, we see a first relation between covering spaces and the fundamental group.

Proposition 2.3.5 Let X be a space and $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space of X . Then the induced homomorphism

$$p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$$

is injective. Moreover, $p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \leq \pi_1(X, x_0)$ consists of all $[f] \in \pi_1(X, x_0)$ such that $\tilde{f}$ is a loop on $\tilde{x}_0$.

Proof Suppose that $[\tilde{f}] \in \ker(p_*)$. Then $[\tilde{f}] : I \rightarrow \tilde{X}$ has the property that $p \circ \tilde{f} \simeq I_{(X, x_0)}$, the trivial loop. This homotopy can be lifted since p has the HLP, which means that $\tilde{f} \simeq I_{(\tilde{X}, \tilde{x}_0)}$ and thus the kernel is trivial.

Clearly if $\tilde{f}$ is a loop on $\tilde{x}_0$ then $f = p \circ \tilde{f}$ lie in $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Conversely if $[f] \in p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ then by HLP there exists a lift of f for which $\tilde{f}$ is a loop. ■

Given the homotopy lifting property and the path lifting property, we have seen that covering maps are special types of spaces that hold these properties. The remainder of this section is to show the partial converse. Given a covering space, what type of maps can the covering space lift? It turns out covering spaces can lift any map provided that the domain is path connected and locally path connected, which we shall see the definition.

Definition 2.3.6 (Locally Path Connected) A topological space X is said to be locally path connected if for every $x \in X$, there exists a neighbourhood basis consisting of open path connected sets.

Theorem 2.3.7 (Lifting Criterion) Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering map and $f : (Y, y_0) \rightarrow (X, x_0)$ a map with Y path connected and locally path connected. Then a lift $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ of f exists:

$$\begin{array}{ccc} & & (\tilde{X}, \tilde{x}_0) \\ & \nearrow \exists! \tilde{f} & \downarrow p \\ (Y, y) & \xrightarrow{f} & (X, x_0) \end{array}$$

if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Moreover, the lift $\tilde{f}$ is unique.

Proof Suppose that $\tilde{f}$ is a lift of f . Then $f = p \circ \tilde{f}$ implies that $f_* = p_* \circ \tilde{f}_*$.

Now let $y \in Y$ and let $\gamma : I \rightarrow Y$ be a path from y_0 to y . Then the path $f \circ \gamma$ has a unique lift $\tilde{f} \circ \gamma$ starting at $\tilde{x}_0$. Define $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ by $\tilde{f}(y) = \tilde{f} \circ \gamma(1)$. We first show that this is well defined. Let $\gamma' : I \rightarrow Y$ be another path from y_0 to y . Then $h = (f \circ \gamma') \cdot (f \circ \gamma)$ is a loop in X at x_0 . By assumption, we have that

$$[h] \in f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

By proposition 2.3.5, $\tilde{h}$ is a loop on $\tilde{x}_0$. By the uniqueness part of the path lifting property, $\tilde{h} = \tilde{f} \circ \gamma' \cdot \tilde{f} \circ \gamma$ so that $\tilde{f} \circ \gamma(1) = \tilde{f} \circ \gamma'(1)$.

Now we show that $\tilde{f}$ is continuous. Let $U \subseteq X$ be an open neighbourhood of $f(y) \in X$. Let $\tilde{U} \subseteq \tilde{X}$ be such that $p|_{\tilde{U}} : \tilde{U} \rightarrow U$ is a homeomorphism with inverse $\iota : U \rightarrow \tilde{U}$. Since Y is locally path connected, we can choose a path connected neighbourhood V of y such that $f(V) \subseteq U$. We want to show that $\tilde{f}|_V = \iota \circ f$ since continuity of ι and f will imply the continuity of $\tilde{f}|_V$ for each V . Let $y' \in V$ be arbitrary. We can find a path γ from y_0 to y and a path η from y to y' where in particular, η lies in V . We now construct a lift of $f(\gamma \cdot \eta) = f(\gamma) \cdot f(\eta)$. From the above paragraph we already have a lift $\tilde{f} \circ \gamma$ of $f \circ \gamma$ ending at $\tilde{f}(y)$. Now since $f \circ \eta$ is entirely contained in U , we have a path $\iota \circ f \circ \eta$ in $\tilde{U} \subseteq \tilde{X}$. This path is in fact a lift of $f \circ \eta$ since

$$p \circ \iota \circ f \circ \eta = f \circ \eta$$

The starting point of $\iota \circ f \circ \eta$ is

$$\iota(f(\eta(0))) = \iota(f(y)) = \tilde{f}(y)$$

This is the ending point of $\widetilde{f \circ \gamma}$ so that

$$f \circ (\widetilde{\gamma \cdot \eta}) = \widetilde{f \circ \gamma} \cdot (\iota \circ f \circ \eta)$$

From this, we have that

$$\begin{aligned} \widetilde{f}(y') &= \left(\widetilde{f \circ \gamma} \cdot (\iota \circ f \circ \eta) \right) (1) \\ &= (\iota \circ f \circ \eta) (1) \\ &= (\iota \circ f)(y') \end{aligned}$$

The continuity of f then follows from the continuity of ι and f . ■

This theorem is rather remarkable since it shows that this topological condition can be characterized entirely with purely algebra.

2.4 Fibers of a Covering Space

The lifting correspondence gives a bijection between the fundamental group and the size of the fibers of the universal covering. This will provide a useful way in calculating the order of the fundamental group.

Definition 2.4.1 (Fibers of a Covering Space) Let X be a space and let $p : \tilde{X} \rightarrow X$ be a covering space. Let $x_0 \in X$. The fiber of p at x_0 refers to the set

$$p^{-1}(x_0) = \{\tilde{x} \in \tilde{X} \mid p(\tilde{x}) = x_0\}$$

We first prove that the fibers of a path connected space X has constant cardinality throughout all of X .

Lemma 2.4.2 Let X be a path connected space and $p : \tilde{X} \rightarrow X$ be a covering space. Then $|p^{-1}(x)|$ is constant throughout all of x .

Proof We will show that for any $x, y \in X$, there is a bijection from $p^{-1}(x)$ to $p^{-1}(y)$. Let $\gamma : I \rightarrow X$ be a path from x to y . For each $\tilde{x} \in p^{-1}(x)$, let $\tilde{\gamma}_{\tilde{x}}$ be the lift of γ starting at $\tilde{x}$. Define the function $\phi : p^{-1}(x) \rightarrow p^{-1}(y)$ by $\tilde{x} \mapsto \tilde{\gamma}_{\tilde{x}}(1)$. Similarly, the inverse path $\bar{\gamma}$ defines a function $\psi : p^{-1}(y) \rightarrow p^{-1}(x)$ by $\tilde{y} \mapsto \bar{\gamma}_{\tilde{y}}(1)$. These functions are inverses of each other since

$$\begin{aligned} (\psi \circ \phi)(\tilde{x}) &= \text{end point of lift of } \bar{\gamma} \text{ starting at } \tilde{\gamma}_{\tilde{x}}(1) \\ &= \text{end point of lift of } \bar{\gamma} \cdot \gamma \text{ starting at } \tilde{x} \\ &= \text{end point of lift of } e_{\tilde{x}} \text{ starting at } \tilde{x} \\ &= \tilde{x} \end{aligned}$$

and similarly we have $\phi(\psi(\tilde{y})) = \tilde{y}$. Thus we conclude. ■

Given that $p^{-1}(x_0)$ is now well defined on varying $x_0 \in X$, we can now formally define the degree of a covering space.

Definition 2.4.3 (Degree of a Covering) Let X be a path connected space and $p : \tilde{X} \rightarrow X$ be a covering space. Define the degree of p to be

$$\deg(p) = |p^{-1}(x)|$$

for any $x \in X$. In this case, we say that p is a $\deg(p)$ -sheeted cover of X .

Lemma 2.4.4 Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering map. Let $[\gamma] \in \pi_1(X, x_0)$ be a loop and $\tilde{\gamma} : I \rightarrow \tilde{X}$ be its unique lift with $\tilde{\gamma}(0) = \tilde{x}_0$. Then the map of sets $\phi : \pi_1(X, x_0) \rightarrow p^{-1}(x_0)$ defined by

$$\phi([\gamma]) = \tilde{\gamma}(1)$$

is well defined.

Proof Suppose that $\gamma \stackrel{\partial}{\simeq} \gamma'$. Then the unique lift of this path homotopy is still a homotopy relative to boundaries. This means that $\tilde{\gamma} \stackrel{\partial}{\simeq} \tilde{\gamma}'$. This implies that $\tilde{\gamma}(1) = \tilde{\gamma}'(1)$ and thus $\phi(\gamma) = \phi(\gamma')$. ■

Definition 2.4.5 (Lifting Correspondence) Let (X, x_0) be a path connected space. Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering map. Define the lifting correspondence to be the map

$$\phi : \pi_1(X, x_0) \rightarrow p^{-1}(x_0)$$

where $\phi([\gamma]) = \tilde{\gamma}(1)$ for $[\gamma] \in \pi_1(X, x_0)$.

Theorem 2.4.6 Let (X, x_0) be a path connected space. Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering map. Denote

$$\phi : \pi_1(X, x_0) \rightarrow p^{-1}(x_0)$$

the lifting correspondence. Then the following are true.

- If $\tilde{X}$ is path connected then ϕ is surjective
- If $\tilde{X}$ is simply connected then ϕ is bijective

Proof

- Let $\tilde{x}_1 \in p^{-1}(x_0)$. Since $\tilde{X}$ is path connected, there is a path $\tilde{\gamma} : I \rightarrow \tilde{X}$ from $\tilde{x}_0$ to $\tilde{x}_1$. This implies that $\gamma = p \circ \tilde{\gamma}$ is a loop in X . Then $[\gamma] \in \pi_1(X, x_0)$ is such that $\phi([\gamma]) = \tilde{x}_1$ and so we conclude.
- Since $\tilde{X}$ is also path connected, we just have to show that ϕ is injective. Suppose that $[f], [g] \in \pi_1(X, x_0)$ is such that $\phi([f]) = \phi([g])$. Then this means that $\tilde{f}(1) = \tilde{g}(1)$. By lemma 1.6.3, we have that $\tilde{f} \stackrel{\partial}{\simeq} \tilde{g}$. This homotopy descends to $[f] = [g]$ and so we conclude. ■

In fact, we can do one step better and look into why injectivity of a path connected space fails for the above theorem. For this, we will need to consider how $\pi_1(X, x_0)$ acts on $p^{-1}(x_0)$ as a group action.

Definition 2.4.7 (Right Action on $p^{-1}(x_0)$) Let X be a space and $x_0 \in X$. Define a right group action $\bullet : p^{-1}(x_0) \times \pi_1(X, x_0) \rightarrow p^{-1}(x_0)$ by the map

$$\tilde{x}_0 \bullet [\gamma] = \tilde{\gamma}(1)$$

where $\tilde{\gamma}$ is the lift of γ starting at $\tilde{x}_0$.

Proposition 2.4.8 Let X be a path connected space and $x_0 \in X$. Let $p : \tilde{X} \rightarrow X$ be a covering space of X . Then $\pi_1(X, x_0)$ acts transitively on $p^{-1}(x_0)$ if and only if $\tilde{X}$ is path connected.

In this case, the stabilizer of $\tilde{x}_0 \in p^{-1}(x_0)$ is given by

$$\text{Stab}_{\pi_1(X, x_0)}(\tilde{x}_0) = \text{im} \left(p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0) \right)$$

Proof Suppose that $\tilde{X}$ is path connected. Then for $\tilde{x}_1, \tilde{x}_2 \in p^{-1}(x_0)$. There exists a path $\tilde{\gamma} : I \rightarrow \tilde{X}$ from $\tilde{x}_1$ to $\tilde{x}_2$. Then $\gamma = p \circ \tilde{\gamma}$ is a loop in $\pi_1(X, x_0)$ such that $\tilde{x}_1 \bullet [\gamma] = \tilde{\gamma}(1) = \tilde{x}_2$.

Now suppose that the right action is transitive. Suppose for a contradiction that $\tilde{x}$ and $\tilde{y}$ lie in different path components of $\tilde{X}$. Since X is path connected, there exists a path $\gamma : I \rightarrow X$ from x to y . Then the lift $\tilde{\gamma} : I \rightarrow \tilde{X}$ start at $\tilde{x}$ and ends at $\tilde{y}_1$ where $p(\tilde{y}_1) = p(\tilde{y})$. By assumption, $\tilde{x}$ and $\tilde{y}_1$ are in the same path component, while $\tilde{y}$ are not in the same path component as both $\tilde{x}$ and $\tilde{y}_1$. But since the right action of $\pi_1(X, x_0)$ is transitive, there exists a $\tilde{\eta} : I \rightarrow \tilde{X}$ from $\tilde{y}$ to $\tilde{y}_1$. This is a contradiction.

Let $\tilde{X}$ be path connected so that the right action is transitive. First let $[\gamma] \in \text{Stab}_{\pi_1(X, x_0)}(\tilde{x}_0)$. Then $\tilde{\gamma}(1) = \tilde{x}_0$. Then this means that $\tilde{\gamma}$ is a loop in $\tilde{X}$ based at $\tilde{x}_0$ so that $[\tilde{\gamma}] \in \text{im}(p_*)$. Now suppose that $[\tilde{\gamma}] \in \pi_1(\tilde{X}, \tilde{x}_0)$. Then $\tilde{\gamma}$ is a lift of $\gamma = p \circ \tilde{\gamma}$, which is a loop in X . Thus we have that $\tilde{x}_0 \bullet \gamma = \tilde{x}_0$ so that $[\gamma] \in \text{Stab}_{\pi_1(X, x_0)}(\tilde{x}_0)$. ■

As one can see, the above proof is reminiscent of the proof of surjectivity of the lifting correspondence. With this proposition, our final goal can be realized easily together with the orbit stabilizer theorem.

Theorem 2.4.9 Let X be a path connected space and $x_0 \in X$. Let $p : \tilde{X} \rightarrow X$ be a path connected covering space of X . Then there is a one to one bijection

$$p^{-1}(x_0) \xleftrightarrow{1:1} \frac{\pi_1(X, x_0)}{p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right)} \quad \text{or} \quad \deg(p) = [\pi_1(X, x_0) : p_* (\pi_1(\tilde{X}, \tilde{x}_0))]$$

between the cosets of $p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right)$ and the fibers of x_0 .

Proof By the orbit stabilizer theorem, we have that

$$[\pi_1(X, x_0) : \text{Stab}_{\pi_1(X, x_0)}(x_0)] = |\text{Orb}_{\pi_1(X, x_0)}(\tilde{x}_0)|$$

Since the right action is transitive, we have that $|\text{Orb}_{\pi_1(X, x_0)}(\tilde{x}_0)| = |p^{-1}(x_0)| = \deg(p)$. Combined with the fact that $\text{Stab}_{\pi_1(X, x_0)}(\tilde{x}_0) = p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right)$, we have that

$$\deg(p) = [\pi_1(X, x_0) : p_* (\pi_1(\tilde{X}, \tilde{x}_0))]$$

and so we conclude. ■

A typical calculation would involve only the case for when $\tilde{X}$ is the universal cover. In this case, by finding the number of fibers of the covering space, we can find out the order of the fundamental group. When the order is low, we can enumerate all possible groups. However notice that we do not yet have a way to calculate $p^{-1}(x)$. This will be done via deck transformations, as they are greatly related to both $p^{-1}(x)$ and $\pi_1(X, x)$.

2.5 The Galois Correspondence of Covering Spaces

We have seen that the fundamental groups under the covering maps provide an image subgroup in the fundamental group of the base space. This gives a mapping

$$\left\{ \begin{array}{c} \text{Covering spaces} \\ \text{over } (X, x_0) \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{Subgroups of} \\ \pi_1(X, x_0) \end{array} \right\}$$

It is natural to ask whether there are equivalence relations on the left so that this is a bijection.

Definition 2.5.1 (Basepoint Equivalence of Covering Spaces) Let $(\tilde{X}_1, \tilde{x}_1)$ and $(\tilde{X}_2, \tilde{x}_2)$ be covering spaces of (X, x_0) . We say that the two pointed covering spaces are base point equivalent if there exists a pointed homeomorphism $h : (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}_2)$ such that the following diagram commutes:

$$\begin{array}{ccc} (\tilde{X}_1, \tilde{x}_1) & \xrightarrow{h} & (\tilde{X}_2, \tilde{x}_2) \\ & \searrow p_1 & \swarrow p_2 \\ & (X, x_0) & \end{array}$$

BP here stands for base point, not to be confused with a similar equivalence 2.6.1 defined on the next subsection.

The following proposition shows that this definition of equivalence on pointed covering spaces works well with our map of sets from covering spaces to subgroups.

Proposition 2.5.2 Let (X, x_0) be a path connected and locally path connected space. Let $p_1 : (\tilde{X}_1, \tilde{x}_1) \rightarrow (X, x_0)$ and $p_2 : (\tilde{X}_2, \tilde{x}_2) \rightarrow (X, x_0)$ be path connected covering spaces of (X, x_0) . Then $(\tilde{X}_1, \tilde{x}_1)$ and $(\tilde{X}_2, \tilde{x}_2)$ are base point equivalent if and only if

$$(p_1)_* \left(\pi_1(\tilde{X}_1, \tilde{x}_1) \right) = (p_2)_* \left(\pi_1(\tilde{X}_2, \tilde{x}_2) \right)$$

In other words, the image subgroup of the push forward of the induced map is the same.

Proof Suppose first that $(\tilde{X}_1, \tilde{x}_1)$ and $(\tilde{X}_2, \tilde{x}_2)$ are equivalent covering spaces via the base point preserving homeomorphism $h : (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}_2)$. Then the relation $p_1 = p_2 \circ h$ implies that $(p_1)_* = (p_2)_* \circ h_*$. Since h_* is an isomorphism, we have that $(p_1)_*$ and $(p_2)_*$ have the same image.

Now suppose that the converse is true. Then all the conditions for lifting criterion are satisfied by p_1 so that we have a lift $\tilde{p}_1$ such that $p_1 = p_2 \circ \tilde{p}_1$. Similarly, we obtain a lift $\tilde{p}_2$ of p_2 such that $p_2 = p_1 \circ \tilde{p}_2$. Now since both $\tilde{p}_2 \circ \tilde{p}_1$ and id are both lifts of p_1 , we must have that they are equal by the unique lifting property. Similarly we have that $\tilde{p}_1 \circ \tilde{p}_2 = \text{id}$ thus we have constructed a homeomorphism between the two coverings. ■

As a consequence of this proposition, we see that simply connected covering spaces are unique up to homeomorphism. Moreover, as an interlude, we will construct and thus prove the existence of the universal covering space under suitably nice conditions. This will in turn allow us to construct corresponding to covering spaces for arbitrary subgroups of the fundamental group in order to complete the Galois correspondence.

Corollary 2.5.3 Any two simply connected covering spaces are base point equivalent.

Proof By definition, simply connected covering spaces have trivial fundamental group and thus the image subgroup of the push forward p_* is also trivial. ■

This space is called the universal covering space and it serves as a keystone in future exchanges between the fundamental group and covering spaces.

Definition 2.5.4 (Universal Covering Space) Let X be a space. The universal covering of X is the simply connected covering space of X that is unique up to homeomorphism, if it exists.

Not every space admits a universal cover. We must impose additional constraints on the space for it to exhibit one. We will prove a weaker version by considering locally simply connectedness instead of semilocally simply connected spaces.

Theorem 2.5.5 (Existence of the Universal Cover) If X is path connected, locally path connected and locally simply connected, then X has a universal cover.

Proof Let $x_0 \in X$. Define the set $\tilde{X}$ to be

$$\tilde{X} = \{[\gamma] \mid \gamma : I \rightarrow X \text{ is a path from } x_0 \text{ to } x \in X\}$$

where $[\gamma]$ is the equivalence class of homotopic paths relative to boundary. Define the covering map $p : \tilde{X} \rightarrow X$ to be $p([\gamma]) = \gamma(1)$. It remains to construct a topology on $\tilde{X}$. Let $\mathcal{U}$ be the collection of simply connected open sets in X . Let $U \in \mathcal{U}$. Write

$$\tilde{U}_{[\gamma]} = \{[\gamma \cdot \eta] \mid \eta \text{ is a path in } U \text{ starting at } \gamma(1)\}$$

What follows is that we will show that the set of all $\tilde{U}_{[\gamma]}$ with varying $[\gamma]$ is a topological basis so that it generates a topology on $\tilde{X}$.

Firstly, $p : \tilde{U}_{[\gamma]}$ is surjective since for any $x \in U$, there exists a path $\eta : I \rightarrow X$ from $\gamma(1) \in U$ to $x \in U$ since U is path connected and $p([\gamma \cdot \eta]) = x$. The map is also injective. Suppose that $p([\gamma \cdot \eta]) = p([\gamma \cdot \eta']) = y \in X$. This means that $[\gamma \cdot \eta]$ and $[\gamma \cdot \eta']$ both have the same endpoints. In particular, both η and η' have the same endpoints. Since U is simply connected, η and η' are homotopic relative to endpoints. Thus $[\gamma \cdot \eta] = [\gamma \cdot \eta']$ represent the same point in $\tilde{X}$ and $[\gamma \cdot \eta] = [\gamma \cdot \eta']$ in $\tilde{U}_{[\gamma]}$.

Now recall the property of covering maps is that $p^{-1}(U)$ must be a disjoint union of homeomorphic pieces of U . We will also prove this. Suppose that $\tilde{U}_{[\gamma]} \cap \tilde{U}_{[\gamma']}$ is non-empty, say $[\gamma'']$ is an element in the intersection. Then we have that $[\gamma''] = [\gamma \cdot \eta]$ for some $\eta : I \rightarrow X$. Elements of $\tilde{U}_{[\gamma']}$ are of the form $[\gamma' \cdot \mu] = [\gamma \cdot \eta \cdot \mu]$. Since $\eta \cdot \mu \subset U$, we have that

$$[\gamma'' \cdot \mu] = [\gamma \cdot (\eta \cdot \mu)] \in \tilde{U}_{[\gamma]}$$

This shows that $\tilde{U}_{[\gamma']} \subseteq \tilde{U}_{[\gamma]}$. We will also prove the reverse inclusion. Elements of $\tilde{U}_{[\gamma]}$ are of the form $[\gamma \cdot \eta]$. We can rewrite this as

$$[\gamma \cdot \eta] = [\gamma \cdot \mu \cdot \bar{\mu} \cdot \eta] = [\gamma' \cdot (\bar{\mu} \cdot \eta)]$$

Since $\bar{\mu} \cdot \eta \subset U$, we have shown that $[\gamma \cdot \eta] \in \tilde{U}_{[\gamma']}$. Thus we have shown that $\tilde{U}_{[\gamma]} = \tilde{U}_{[\gamma']}$. A similar argument shows that $\tilde{U}_{[\gamma]} = \tilde{U}_{[\gamma']}$. This completes the proof that p is a covering map.

It remains to show that $\{\tilde{U}_{[\gamma]} \mid [\gamma] \in \tilde{X}\}$ is a basis of $\tilde{X}$. Let $[\gamma] \in \tilde{X}$. Then $\tilde{U}_{[\gamma]}$ clearly contains $[\gamma]$. Thus the collection covers $\tilde{X}$. Suppose that we are given $\tilde{U}_{[\gamma]}$ and $\tilde{V}_{[\gamma']}$ with non-empty intersection, say $[\gamma'']$ lies in it. By a similar argument as above, we have that $\tilde{U}_{[\gamma]} = \tilde{U}_{[\gamma']}$ and $\tilde{V}_{[\gamma]} = \tilde{V}_{[\gamma']}$. Since $\mathcal{U}$ is a basis of X , there exists some $W \in \mathcal{U}$ such that $W \subset U \cap V$ and containing $p(\gamma'') = \gamma''(1)$. Then we have that

$$[\gamma''] \in \tilde{W}_{[\gamma'']} \subset \tilde{U}_{[\gamma'']} \cap \tilde{V}_{[\gamma']}$$

and the collection $\tilde{U}_{[\gamma]}$ forms a basis for $\tilde{X}$.

We now show that p is continuous. Notice that since $p : \tilde{U}_{[\gamma]} \rightarrow U$ gives a bijection between subsets $\tilde{V}_{[\gamma']} \subseteq \tilde{U}_{[\gamma]}$ and $V \subseteq U$. This is because if $\tilde{V}_{[\gamma']} \subseteq \tilde{U}_{[\gamma]}$ then $p(\tilde{V}_{[\gamma']}) = V$ and if $V \subseteq U$, then $p^{-1}(V) \cap \tilde{U}_{[\gamma]}$ is $\tilde{V}_{[\gamma']}$ where $\gamma' \in \tilde{U}_{[\gamma]}$ with endpoints in V since $\tilde{V}_{[\gamma']} \subset U_{[\gamma']} = \tilde{U}_{[\gamma]}$ and $\tilde{V}_{[\gamma']}$ maps onto V by the injection p . This means that p is a homeomorphism.

It remains to show that $\tilde{X}$ is simply connected. Choose a basepoint $\tilde{x}_0$ for $\tilde{X}$. Let $\tilde{x}_0 = [e_{x_0}]$. Notice that $p([e_{x_0}]) = e_{x_0}(1) = x_0$. We know that $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$ is injective. It remains to

show that the image of p_* is trivial. Recall that $[\gamma] \in \text{im}(p_*)$ if and only if γ lifts to a loop in $(\tilde{X}, \tilde{x}_0)$. So we have to show that a loop γ in (X, x_0) lifts to a loop in $(\tilde{X}, \tilde{x}_0)$ homotopic to the constant loop e_{x_0} . The definition of $\tilde{X}$ implies that a loop in $\tilde{X}$ is a path of paths. Let $\gamma : I \rightarrow X$ be a loop in X based at x_0 . Write

$$\gamma_t(s) = \begin{cases} \gamma(s) & \text{if } 0 \leq s \leq t \\ \gamma(t) & \text{if } t \leq s \leq 1 \end{cases}$$

so that for each fixed t , $\gamma_t(s)$ is the path in X that starts by tracing γ until $s = t$, then it stays at $\gamma(t)$ until $s = 1$. Now $t \mapsto [\gamma_t]$ is a path in $\tilde{X}$. Since $p(\gamma_t) = \gamma_t(1) = \gamma(1) = \gamma(t)$, the path $[\gamma_t]$ in $\tilde{X}$ lifts the path γ in X starting at $[\gamma_0] = [\gamma_0(s)] = [e_{x_0}(s)] = \tilde{x}_0$. But $t \mapsto [\gamma_t]$ being a loop in $\tilde{X}$ implies that $[\gamma_0] = [\gamma_1]$ and that $[\gamma_1] = [e_{x_0}]$. Now $\gamma_1(s) = \gamma(s)$ implies that $\gamma(s) \stackrel{\partial}{\simeq} e_{x_0}(s)$ and that $\gamma(s)$ is homotopically trivial.

Using the universal cover, we can find all the other covering spaces of X .

Theorem 2.5.6 If X is path connected, locally path connected and locally simply connected and H is a subgroup of $\pi_1(X, x_0)$, then there is a unique covering space $p : \tilde{X} \rightarrow X$ with $p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = H$.

We can finally state the Galois theory for covering spaces in its full form.

Theorem 2.5.7 (Galois Correspondence for Covering Spaces (Pointed)) Let (X, x_0) be a pointed topological space that is path connected, locally path connected and locally simply connected. Then there is a bijection between path connected covering spaces up to base point equivalence and subgroups of $\pi_1(X, x_0)$.

$$\left\{ \begin{array}{c} \text{Covering spaces} \\ \text{over } (X, x_0) \end{array} \right\} / \cong \xleftrightarrow{1:1} \left\{ \begin{array}{c} \text{Subgroups of} \\ \pi_1(X, x_0) \end{array} \right\}$$

where $(Y, y_0) \cong (Z, z_0)$ if and only if they are base point equivalent covering spaces over (X, x_0) in the sense of ??.

The bijection sends $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ to the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Proof By 2.5.2, this map is well defined. Moreover, 2.5.2 also shows that this map is injective. By 2.5.6, this map is surjective and so we conclude. ■

This bijection is more over inclusion reversing in the sense that if K and H are subgroups of $\pi_1(X, x_0)$ with their corresponding covering maps $\tilde{X}_K$ and $\tilde{X}_H$ respectively, then K is a subgroup of H if and only if $\tilde{X}_K$ is a covering space of $\tilde{X}_H$. We will not show the proof here.

Inclusion reversing bijections of this type are often called Galois correspondence due to his work on the inclusion reversing bijection between field extensions and subgroups of the Galois group.

2.6 Unpointed Equivalent Coverings and Conjugacy Classes

The final section will improve on the Galois corresponding by considering equivalent covering spaces without considering where the base point is sent. This will introduce a new equivalence relation on the equivalence classes of basepoint preserving isomorphisms of covering spaces. Indeed if (Y, y_0) and (Z, z_0) are isomorphic pointed covering spaces, (Y, y_0) and (Z, z_1) for z_0, z_1 lying in the same fiber should also be isomorphic since (Z, z_0) and (Z, z_1) differ by a deck transformation, which is a homeomorphism from Z to itself.

Definition 2.6.1 (Equivalent Covering Spaces) Let $\tilde{X}_1$ and $\tilde{X}_2$ be covering spaces of X . We say that the two pointed covering spaces are equivalent if there exists a homeomorphism $h : \tilde{X}_1 \rightarrow \tilde{X}_2$ such that the following diagram commutes:

$$\begin{array}{ccc} \tilde{X}_1 & \xrightarrow{h} & \tilde{X}_2 \\ & \searrow p_1 & \swarrow p_2 \\ & X & \end{array}$$

We first prove a mild result in within a single covering space to motivate the introduction of conjugacy classes in the Galois correspondence.

Proposition 2.6.2 Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a path connected covering space and $\tilde{x}_0 \in p^{-1}(x_0)$. Then there is a bijection

$$\left\{ \begin{array}{l} \text{Conjugacy Classes} \\ \text{of } p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \end{array} \right\} \xleftrightarrow{1:1} p^{-1}(x_0)$$

This means that the subgroups $p_*(\pi_1(\tilde{X}, \tilde{x}_1))$ for $\tilde{x}_1 \in p^{-1}(x_0)$ are precisely a conjugacy class of the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

In particular, the bijection is given as follows: For $\tilde{x}_1 \in p^{-1}(x_0)$, let $\bar{f} : I \rightarrow \tilde{X}$ be a path from $\tilde{x}_0$ to $\tilde{x}_1$. Write $f = p \circ \bar{f}$. Then

$$p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = [f] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_1)) \cdot [\bar{f}]$$

Proof Define a map $\tilde{x}_1 \mapsto [f] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_1)) \cdot [\bar{f}]$. We first show that this is well defined. Namely that $[f] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_1)) \cdot [\bar{f}]$ is indeed $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Let $[\alpha] \in p_*(\pi_1(\tilde{X}, \tilde{x}_1))$. Then by injectivity of p_* , there exists a unique $[\tilde{\alpha}] \in \pi_1(\tilde{X}, \tilde{x}_1)$ such that $p_*([\tilde{\alpha}]) = [\alpha]$. Notice that $[\bar{f} \cdot \alpha \cdot \bar{f}] \in \pi_1(\tilde{X}, \tilde{x}_0)$. Passing through p_* , we have that

$$\begin{aligned} p_*([\bar{f} \cdot \alpha \cdot \bar{f}]) &= [(p \circ \bar{f}) \cdot (p \circ \alpha) \cdot (p \circ \bar{f})] \\ &= [f \cdot \alpha \cdot \bar{f}] \end{aligned}$$

Thus we have that $[f] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_1)) \cdot [\bar{f}] \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Now suppose that $[\beta] \in p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Notice that $\bar{f}$ is a path from $\tilde{x}_1$ to $\tilde{x}_0$. So by a similar argument as above, we have that $[\bar{f}] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \cdot [f] \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_1))$. This implies that

$$p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subseteq [f] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_1)) \cdot [\bar{f}]$$

and so we are done.

It remains to show that this map has an inverse. Suppose that $p_*(\pi_1(\tilde{X}, \tilde{x}_1))$ is conjugate to $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Then there exists $[\alpha] \in \pi_1(X, x_0)$ such that

$$p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = [\alpha] \cdot p_*(\pi_1(\tilde{X}, \tilde{x}_1)) \cdot [\bar{\alpha}]$$

By the path lifting property of covering maps, and uniqueness of lifts, there exists a unique lift $\tilde{\alpha}$ of α with starting point $\tilde{x}_0$. Moreover $\tilde{x}_1 = \tilde{\alpha}(1)$ lies in $p^{-1}(x_0)$. Thus we have constructed an inverse. ■

Keen readers would have realized that the first half of the proof involves showing that the following diagram:

$$\begin{array}{ccc}
\pi_1(\tilde{X}, \tilde{x}_0) & \xrightarrow{[\gamma] \mapsto [\tilde{f} \cdot \gamma \cdot \tilde{f}]} & \pi_1(\tilde{X}, \tilde{x}_1) \\
\downarrow p_* & & \downarrow p_* \\
p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right) & \xrightarrow{[\eta] \mapsto [\tilde{f} \cdot \eta \cdot \tilde{f}]} & p_* \left(\pi_1(\tilde{X}, \tilde{x}_1) \right)
\end{array}$$

is commutative.

The proposition motivates the following unpointed equivalence covering characterization similar to 2.5.2

Proposition 2.6.3 Let X be a path connected and locally path connected space. Let $p : \tilde{X} \rightarrow X$ and $p' : \tilde{X}' \rightarrow X$ be path connected covering spaces of X . Then $\tilde{X}$ and $\tilde{X}'$ are equivalent if and only if there exists $\tilde{x}_0 \in p^{-1}(x_0)$ and $\tilde{x}'_0 \in (p')^{-1}(x_0)$ such that the subgroups

$$p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right) \quad \text{and} \quad p'_* \left(\pi_1(\tilde{X}', \tilde{x}'_0) \right)$$

are conjugate.

Proof Suppose that $h : \tilde{X} \rightarrow \tilde{X}'$ is an equivalence of covering spaces. Let $x_0 \in X$ be arbitrary. Let $\tilde{x}_0 \in p^{-1}(x_0)$. Let $\tilde{x}'_1 = h(\tilde{x}_0)$. Then we can regard h as a base point preserving map so that we can apply 2.5.2 to obtain

$$p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right) = p'_* \left(\pi_1(\tilde{X}', \tilde{x}'_1) \right)$$

By 2.6.2, we have that

$$p_* \left(\pi_1(\tilde{X}', \tilde{x}'_1) \right) \quad \text{and} \quad p'_* \left(\pi_1(\tilde{X}', \tilde{x}'_0) \right)$$

are conjugate so we are done.

Suppose now that we are given two conjugate subgroups $p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right)$ and $p'_* \left(\pi_1(\tilde{X}', \tilde{x}'_0) \right)$. By the above proposition, there exists a $\tilde{x}'_1$ such that

$$p'_* \left(\pi_1(\tilde{X}', \tilde{x}'_1) \right) = p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right)$$

By 2.5.2, we obtain a base point preserving map $h : (\tilde{X}, \tilde{x}_0) \rightarrow (\tilde{X}', \tilde{x}'_1)$. In particular, by forgetting the base point, we obtain an equivalence $h : \tilde{X} \rightarrow \tilde{X}'$ and so we conclude. ■

Finally, we obtain the Galois theory for general covering spaces without the need to consider base-points.

Theorem 2.6.4 (Galois Theory for Covering Spaces (Unpointed)) Let X be a topological space that is path connected, locally path connected and locally simply connected. Then there is a one to one bijection between path connected covering spaces up to equivalence (2.6.1) and subgroups of $\pi(X, x_0)$ up to conjugacy classes. (Choice of x_0 does not matter since path connectedness)

$$\frac{\left\{ \begin{array}{c} \text{Covering spaces} \\ \text{over } X \end{array} \right\}}{\cong} \xleftrightarrow{1:1} \frac{\left\{ \begin{array}{c} \text{Subgroups of} \\ \pi(X, x_0) \end{array} \right\}}{\text{Conjugacy}}$$

where $Y \cong Z$ if and only if they are equivalent covering spaces over X in the sense of 2.6.1.

The map sends $p : \tilde{X} \rightarrow X$ to the conjugacy class $[p_* \left(\pi_1(\tilde{X}, \tilde{x}_0) \right)]$.

Proof By 2.6.3, this map is well defined. Also by 2.6.3, this map is injective. By 2.5.6, this map is surjective and so we conclude. ■

Similar to the pointed version of the Galois correspondence, this bijection is moreover inclusion reversing, which we will not show here.

3 Computing the Fundamental Group

While the theory of covering spaces gives us some level of computation of fundamental groups, we will need a major theorem on patching fundamental groups of subspaces called the Seifert-van Kampen theorem.

The first section we also show that just from the fundamental group of the circle, we can already deduce some remarkable consequences such as Brouwer's fixed point theorem and the fundamental theorem of algebra.

3.1 The Fundamental Group of the Circle

In this section we compute our first fundamental group through the use of covering space and path lifting properties. The proof for the fundamental group of the circle reproduces important concepts of the proofs of the covering space, instead of directly applying all the consequences of the HLP. This means that the proof bears some resemblance to the more abstract proof of the HLP, which might also be a good exercise in trying to understand the proofs in the covering space.

Theorem 3.1.1 There is a group isomorphism

$$\pi_1(S^1, 1) \cong \mathbb{Z}$$

given as follows. For each $n \in \mathbb{Z}$, $\phi : \mathbb{Z} \rightarrow \pi_1(S^1, 1)$ sends n to $[u_n : I \rightarrow S^1]$ where $u_n(t) = e^{2\pi i n t}$.

Proof Define the map $p : \mathbb{R} \rightarrow S^1$ by $s \mapsto e^{2\pi i s}$. Notice that this is a covering map. Indeed, for each $x \in S^1 \setminus \{1\}$, $S^1 \setminus \{1\}$ is an open neighbourhood of x for which

$$p^{-1}(S^1 \setminus \{1\}) = \coprod_{n \in \mathbb{Z}} (n, n+1)$$

and on each component of the disjoint union p gives a homeomorphism. Hence $p : \mathbb{R} \rightarrow S^1$ is a covering map.

By the path lifting property, $u_n : I \rightarrow S^1$ has a lift $\widetilde{u}_n : I \rightarrow \mathbb{R}$, given by $t \mapsto nt$. Now define $\tau_m : \mathbb{R} \rightarrow \mathbb{R}$ by $\tau_m(s) = s + m$. I claim that $\widetilde{u_m + n}$ is homotopic to $\widetilde{u_m} \cdot (\tau_m \circ \widetilde{u_n})$ relative boundary. Define a map $H : I \times I \rightarrow \mathbb{R}$ by

$$H(t, s) = (1-s)\widetilde{u_{m+n}}(t) + s(\widetilde{u_m} \cdot (\tau_m \circ \widetilde{u_n}))(t)$$

It is easy to see that

- $H(t, 0) = \widetilde{u_{m+n}}(t)$
- $H(t, 1) = \widetilde{u_m} \cdot (\tau_m \circ \widetilde{u_n})(t)$
- $H(0, s) = (1-s) \cdot 0 + s \cdot 0 = 0$
- $H(1, s) = (1-s) \cdot (m+n) + s \cdot (m+n) = m+n$

So that H is a well defined homotopy relative to boundary.

To show that ϕ is a well defined group homomorphism, we calculate that

$$\begin{aligned} [u_{m+n}] &= [p \circ \widetilde{u_{m+n}}] && \text{(Lift of } u) \\ &= [p \circ (\widetilde{u_m} \cdot (\tau_m \circ \widetilde{u_n}))] && \text{(Homotopies descend)} \\ &= [(p \circ \widetilde{u_m}) \cdot (p \circ \tau_m \circ \widetilde{u_n})] && \text{(Paths descend)} \\ &= [p \circ \widetilde{u_m}] \cdot [p \circ \tau_m \circ \widetilde{u_n}] && \text{(Concatenation is well defined up to homotopy)} \\ &= [p \circ \widetilde{u_m}] \cdot [p \circ \widetilde{u_n}] && (p \circ \tau_m = p) \\ &= [u_m] \cdot [u_n] && \text{(Lift of } u) \end{aligned}$$

Thus ϕ is indeed a group homomorphism.

We now show that ϕ is surjective. Suppose that $[v] \in \pi_1(S^1, 1)$. By the path lifting property of p , there exists a unique lift $\tilde{v} : I \rightarrow \mathbb{R}$ such that $v = p \circ \tilde{v}$ and $\tilde{v}(0) = 0$. In particular, we have that

$$\tilde{v}(1) \in p^{-1}(1) = \{t \in \mathbb{R} \mid e^{2\pi it} = 1\} = \mathbb{Z}$$

so that $\tilde{v}(1)$ is an integer, say $\tilde{v}(1) = k \in \mathbb{Z}$. I claim that $\tilde{v}$ and $\tilde{u}_k$ are homotopic. Define a homotopy $H : I \times I \rightarrow \mathbb{R}$ by

$$H(t, s) = (1 - s)\tilde{v}(t) + s\tilde{u}_k(t)$$

It is clear that it satisfies the boundary conditions so that H is a homotopy from $\tilde{v}$ to $\tilde{u}_k$ fixing end points. Since p is a covering map, by homotopy descend we have that $[v] = [\tilde{u}_k]$ implies

$$[v] = [p \circ \tilde{v}] = [p \circ \tilde{u}_k] = [u_k]$$

Hence ϕ is surjective.

We also need to show that ϕ is injective. Let $[u_n] \in \pi_1(S^1, 1)$. Suppose that e_1 be the constant loop on $1 \in S^1$. I claim that if $[u_n] = [e_1]$ then $n = 0$. If $[u_n] = [e_1]$, then there exists a homotopy relative boundary $H : I \times I \rightarrow S^1$ such that $H(t, 0) = u_n(t)$ and $H(t, 1) = e_1(t) = 1$ and $H(0, s) = H(1, s) = 1$. Recall that the path $\tilde{u}_n : I \rightarrow \mathbb{R}$ defined by $t \mapsto nt$ is a lift of u_n . Since p is a covering map, it has the homotopy lifting property. This means that there exists a homotopy $\tilde{H} : I \times I \rightarrow \mathbb{R}$ such that the following diagram commutes:

$$\begin{array}{ccc} I \times \{0\} & \xrightarrow{\tilde{u}_n} & \mathbb{R} \\ \downarrow & \nearrow \exists \tilde{H} & \downarrow p \\ I \times I & \xrightarrow{H} & S^1 \end{array}$$

Now since $p \circ \tilde{H} = H$, we have that

$$\tilde{H}(0, s), \tilde{H}(1, s) \in p^{-1}(1) = \{t \in \mathbb{R} \mid e^{2\pi it} = 1\} = \mathbb{Z}$$

for $s \in I$. But since H is continuous in s and $\mathbb{Z}$ is a discrete subset of $\mathbb{R}$, we must have that $\tilde{H}(0, s) = \tilde{H}(0, 0) \in \mathbb{Z}$ is constant for $s \in I$, and $\tilde{H}(1, s) = \tilde{H}(1, 0) \in \mathbb{Z}$ is constant for $s \in I$. But by definition of the lift of the homotopy, we have $\tilde{H}(0, 0) = 0$. Similarly, $\tilde{H}(1, 0) = n$. At the same time, we have

$$\tilde{H}(t, 1) \in p^{-1}(1) = \{t \in \mathbb{R} \mid e^{2\pi it} = 1\} = \mathbb{Z}$$

so by a similar argument, we conclude that $\tilde{H}(1, 1) = \tilde{H}(t, 1) = \tilde{H}(0, 1)$ for any $t \in I$. Then we have

$$\begin{aligned} \tilde{H}(0, 0) &= 0 \\ \tilde{H}(0, s) &= 0 & (\text{for any } s \in I) \\ \tilde{H}(0, 1) &= 0 \\ \tilde{H}(t, 1) &= 0 & (\text{for any } t \in I) \\ \tilde{H}(1, 1) &= 0 \\ \tilde{H}(1, s) &= 0 & (\text{for any } s \in I) \\ \tilde{H}(1, 0) &= 0 \\ \tilde{u}_n(1) &= 0 \\ n &= 0 \end{aligned}$$

Hence ϕ is injective. ■

Example 3.1.2 (Fundamental Group of T^n) The fundamental group of the n -torus is

$$\pi_1(T^n, x_0) \cong \pi_1(S^1, 1) \times \cdots \times \pi_1(S^1, 1)$$

the product of n fundamental groups of the circle for any $x_0 \in T^n$.

Proof Since T^n is path connected, the choice of base point does not matter. By the product rule of the fundamental group, we conclude. ■

There are a number of important consequences deducible from the fundamental group of circle, via comparing it to other fundamental groups.

Proposition 3.1.3 The disc D^2 does not retract to $S^1 \subset D^2$.

Proof If there exists a retract $f : D^2 \rightarrow S^1$ then $\pi_1(f)$ is a surjective homomorphism. But $\pi_1(D^2, 1) \cong \{0\}$ and $\pi_1(S^1, 1) \cong \mathbb{Z}$ which is a contradiction. ■

Brouwer's fixed point theorem is a powerful theorem showing that every map from D^n to D^n has a fixed point. During his time there was not the notion of the fundamental group. Now with the fundamental group, we can treat the case of $n = 2$ very easily.

Proposition 3.1.4 (Brouwer's Fixed Point Theorem) Every continuous map $f : D^2 \rightarrow D^2$ has a fixed point.

Proof Suppose for a contradiction that $f : D^2 \rightarrow D^2$ has no fixed points. Then for every x and $f(x)$, there is a unique line $L_x(t) = (1-t)f(x) + tx$ joining x and $f(x)$. The line intersects with S^1 in two points. Let $r(x)$ be the one where $t > 0$. Then clearly $r : D^2 \rightarrow S^1$ is a function such that $r(x) = x$ for $x \in S^1$.

I claim that this function is continuous and thus is a retract. We want $|L_x(t)| = 1$ which means that

$$((1-t)f(x) + tx)^2 = 1$$

Clearly the positive solution of this quadratic depends continuously on x and $f(x)$ which is also continuous. This means that r is indeed a retract. This contradicts ?? saying that D^2 does not retract to S^1 thus we are done. ■

To show the more general Brouwer's fixed point theorem, we make use of homology in the sequel of these notes.

The Borsuk-Ulam theorem for the 2-dimensional case is a result of Brouwer's fixed point theorem. Note that it is in general not true if we replace S^2 to any other space, such as the torus.

Proposition 3.1.5 (Borsuk-Ulam Theorem) Let $f : S^2 \rightarrow \mathbb{R}^2$ be a map. Then there exists a point $x \in S^2$ such that $f(x) = f(-x)$.

Proof Suppose that the result is false. Then we can define a map $g : S^2 \rightarrow S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

It is clear that $g(-x) = -g(x)$. Define the upper hemisphere

$$U = \{(x, y, z) \in S^2 \mid z \geq 0\}$$

and a map $q : S^1 \times I \rightarrow U$ by

$$q(e^{i\theta}, t) = (t \cos(\theta), t \sin(\theta), \sqrt{1-t^2})$$

Then $q(e^{-\theta}, 0) = (0, 0, 1)$ the constant map and $q(e^{i\theta}, 1) = (\cos(\theta), \sin(\theta), 0)$ embeds the circle S^1 into the equator. Now consider the homotopy $H : S^1 \times I \rightarrow S^1$ defined by $H(e^{i\theta}, t) = g \circ q$. Then

$H(e^{i\theta}, 0) = (g \circ q)(e^{i\theta}, 0)$ is a constant map and $H(e^{i\theta}, 1) = (g \circ q)(e^{i\theta}, 1)$ is an odd map since both g and $q(-, 1)$ is odd. Thus we have that $g \circ q(-, 1) : S^1 \rightarrow S^1$ is null homotopic.

Since S^1 is path connected, any two constant maps on S^1 are homotopic and so in particular, suppose that $h = g \circ q(-, 1) : S^1 \rightarrow S^1$ is homotopic to e_1 by $F : S^1 \times I \rightarrow S^1$. Let $k(x) = \frac{h(x)}{h(1)}$. Clearly we have that $k(-x) = -k(x)$ and $k(1) = 1$. Thus we can define a loop $\gamma : I \rightarrow S^1$ by $\gamma(s) = k(e^{2\pi is})$. Since $e^{i\pi} = -1$, we get that

$$\gamma\left(s + \frac{1}{2}\right) = k(e^{2\pi is + \pi i}) = -\gamma(s)$$

for $s \in [0, 1/2]$. In particular, we have that $\gamma(\frac{1}{2}) = -1$. By the path lifting property, we obtain a path $\tilde{\gamma} : I \rightarrow \mathbb{R}$ with $\tilde{\gamma}(0) = 0$ and $\tilde{\gamma}(\frac{1}{2}) = n + \frac{1}{2}$ for some $n \in \mathbb{Z}$. We want to show that $\tilde{\gamma}(1) = 2n + 1 \in p^{-1}(-1)$.

Consider two paths $\alpha, \beta : I \rightarrow \mathbb{R}$ defined by $\alpha(s) = n + \frac{1}{2} + \tilde{\gamma}(\frac{s}{2})$ and $\beta(s) = \tilde{\gamma}(\frac{s+1}{2})$. We then have that for $p : \mathbb{R} \rightarrow S^1$ the covering map,

$$(p \circ \alpha)(s) = -\gamma\left(\frac{s}{2}\right) = \gamma\left(\frac{s+1}{2}\right) = p \circ \beta$$

and $\alpha(0) = \beta(0) = n + \frac{1}{2}$. By the uniqueness of lifts, it follows that $\alpha = \beta$ and hence $2n + 1 = \alpha(1) = \beta(1) = \tilde{\gamma}(1)$.

Recall the homotopy F from h to e_1 . We can use this to define a homotopy $G : I \times I \rightarrow S^1$ by $G(s, t) = \frac{F(e^{2\pi is}, t)}{F(1, t)}$ where division is over the complex numbers. This is a homotopy from γ to e_1 since $G(s, 0) = \gamma(s)$, $G(s, 1) = 1$ and $G(0, t) = G(1, t) = 1$. Since $\gamma \simeq e_1$, the homotopy ascends uniquely $\tilde{\gamma} \simeq \tilde{e}_1$. But any lifts of a constant loop to a covering is also constant, thus $\tilde{\gamma}$ is null homotopic. This is a contradiction since $\tilde{\gamma}(0) = 0$ and $\tilde{\gamma}(1) = 2n + 1$. Thus we conclude. ■

3.2 Seifert-van Kampen Theorem

Seifert-van Kampen's theorem allows to decompose large spaces into smaller spaces for computation of the fundamental group. We calculate the fundamental group of the smaller spaces and glue it back up.

Recall the amalgamated product of two groups.

Theorem 3.2.1 (Baby Seifert-van Kampen Theorem) Let X be a space such that $A, B \subseteq X$ covers X . Let $A \cap B$ be path connected and non-empty. Let $x_0 \in A \cap B$. Then there exists a group isomorphism

$$\Psi : \pi_1(A, x_0) *_{\pi_1(A \cap B, x_0)} \pi_1(B, x_0) \rightarrow \pi_1(X, x_0)$$

from the amalgamated product such that the following diagram commutes:

$$\begin{array}{ccccc} & \pi_1(A, x_0) & & & \\ & \nearrow i_1 & & \searrow j_1 & \\ \pi_1(A \cap B, x_0) & & \pi_1(A, x_0) *_{\pi_1(A \cap B, x_0)} \pi_1(B, x_0) & \xrightarrow{\exists \Psi} & \pi_1(X, x_0) \\ & \searrow i_2 & & \nearrow j_2 & \\ & \pi_1(B, x_0) & & & \end{array}$$

where i_1, i_2 are induced by the inclusions $A \cap B$ to A and B respectively and j_1, j_2 are induced by the inclusions A, B to X respectively.

Category theorists can realize that this is a pushout diagram of the inclusions $\pi_1(A \cap B, x_0) \rightarrow \pi_1(A, x_0), \pi_1(B, x_0)$.

Example 3.2.2 (Fundamental Group of S^n)

Let $n \geq 2$ be an integer and $x_0 \in S^n$. Then $\pi_1(S^n, x_0) \cong \{0\}$.

Proof

Notice that the choice of basepoint does not matter since for $n \geq 2$, S^n is path connected. Let $U = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_{n+1} > 0 - \epsilon\}$ be the upper hemisphere of S^n and $V = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_{n+1} < 0 + \epsilon\}$ be the lower hemisphere of S^n for $\epsilon > 0$ a small number. It is clear that $U \cup V = S^n$ and $U \cap V \simeq \mathbb{R}^n \setminus \{0\}$ is homotopic equivalent to S^{n-1} . But U and V are contractible. By the Seifert-van Kampen theorem, we obtain

$$\pi_1(S^n, x_0) \cong \{1\} *_{\pi_1(S^{n-1}, s)} \{1\} \cong \{1\}$$

and so we conclude. ■

We will see how the lifting correspondence helps us deduce the fundamental group of the projective space.

Example 3.2.3 (Fundamental Group of Projective Space)

For $n \geq 2$, the fundamental group of $\mathbb{RP}^n$ is

$$\pi_1(\mathbb{RP}^n, x_0) \cong \mathbb{Z}/2\mathbb{Z}$$

for any base point $x_0 \in \mathbb{RP}^n$.

Proof

Recall that $\mathbb{RP}^n$ is the set of all lines passing through 0 in $\mathbb{R}^{n+1}$. Each line intersects with S^n in exactly two points where $|x| = 1$. Thus S^n is a 2-sheeted covering of $\mathbb{RP}^n$. But $\pi_1(S^n, x) \cong \{1\}$ which means that S^n is the universal covering of $\mathbb{RP}^n$. By 2.4.6, we have that $\pi_1(\mathbb{RP}^n, x)$ is of order 2 since $p : S^n \rightarrow \mathbb{RP}^n$ is a 2-sheeted cover. But there is only one unique group of order 2 up to isomorphism and so we conclude. ■

Corollary 3.2.4 Let X be a space such that $A, B \subseteq X$ covers X . If $A \cap B$ is simply connected, then for any $x_0 \in A \cap B$, there is an isomorphism

$$\pi_1(X, x_0) \cong \pi_1(A, x_0) * \pi_1(B, x_0)$$

between the fundamental group of X and the free product.

Proof Immediate by 3.2.1. ■

The fundamental group becomes the free product instead of the amalgamated product because $A \cap B$ is simply connected. Then $\pi_1(A \cap B, x_0)$ being trivial means that there is no relation on gluing $\pi_1(A, x_0)$ and $\pi_1(B, x_0)$.

The full version of the theorem is as follows.

Theorem 3.2.5 Let X be a space. Let $\{U_\alpha \mid \alpha \in I\}$ be a path connected open cover of X such that $x_0 \in U_\alpha$ for all α . Denote

$$\Psi : *_{\alpha} \pi_1(U_\alpha) \rightarrow \pi_1(X, x_0)$$

be the homomorphism induced by the inclusions $U_\alpha \hookrightarrow X$. Then the following are true.

- If each $U_\alpha \cap U_\beta$ is path connected, then Ψ is surjective.
- If each $U_\alpha \cap U_\beta \cap U_\gamma$ is path connected, then the kernel

$$\ker(\Psi) = N \left(\left\{ \iota_{\alpha\beta, \alpha}(\omega) (\iota_{\alpha\beta, \beta}(\omega))^{-1} \mid \alpha, \beta \in I \right\} \right)$$

is the normalizer generated by the maps $\iota_{\alpha\beta, \alpha} : \pi_1(U_\alpha \cap U_\beta, x_0) \rightarrow \pi_1(U_\alpha)$ induced by the inclusions $U_\alpha \cap U_\beta \hookrightarrow U_\alpha$.

Proof Note that Ψ is induced by the inclusions $i_\alpha : U_\alpha \hookrightarrow X$. Indeed the collection of induced homomorphisms $(i_\alpha)_* : \pi_1(U_\alpha, x_0) \rightarrow \pi_1(X, x_0)$ uniquely extends to a group homomorphism

$$*_\alpha \pi_1(U_\alpha) \rightarrow \pi_1(X, x_0)$$

which is where Ψ comes from. ■

Reminiscing the baby Seifert-van Kampen theorem, the above general isomorphism

$$\frac{*_\alpha \pi_1(U_\alpha, x_0)}{\ker(\Psi)} \cong \pi_1(X, x_0)$$

is precisely the amalgamated product of a collection of groups.

3.3 Wedge Sum of Spaces

Definition 3.3.1 (Wedge Sum of Spaces) Let $\{(X_i, x_i) \mid i \in I\}$ be a family of pointed spaces. Define their wedge sum to be the quotient space

$$\bigvee_{i \in I} (X_i, x_i) = \frac{\coprod_{i \in I} X_i}{\coprod_{i \in I} \{x_i\}}$$

Lemma 3.3.2

Let (X, x_0) and (Y, y_0) be pointed spaces. If the base point of $(X, x_0) \vee (Y, y_0)$ has a simply connected neighbourhood, then there is an isomorphism

$$\pi_1((X, x_0) \vee (Y, y_0)) \cong \pi_1(X, x_0) * \pi_1(Y, y_0)$$

given by Seifert-van Kampen theorem.

Proof

Follows directly from the theorem. ■

4 The Higher Homotopy Groups

The journey of Algebraic Topology began with the fundamental group, where we assigned a group to every space functorially. The notion of fundamental group heavily involves the notion of homotopy and therefore is heavily related to the notion of homotopy. However, one realizes that even with Seifert-van Kampen theorem and the theory of covering spaces, it is not easy to compute the fundamental group of a space. This is partly, but not wholly due to the fundamental group is in general not abelian. If we instead work in an abelian setting, one is able to distinguish two non-isomorphic groups simply by analysing the torsion subgroups. Therefore we refine the concept of the fundamental group and procured the notion of homology and cohomology. Both functorial invariants now produce graded abelian groups for each space, one for each dimension $n \in \mathbb{N}$. In the case of cohomology, there is a canonical ring structure on cohomology that interacts with the topology of the underlying space.

Now we turn to the final main invariant of topological spaces. The homotopy groups $\pi_n(X, x_0)$ serves as both a generalization of the fundamental group $\pi_1(X, x_0)$ in higher dimensions and a homotopic analogue to homology via the Hurewicz homomorphism

$$h : \pi_n(X) \rightarrow H_n(X)$$

It is a strong invariant that is closely related to the notion of homotopy, all the while having mostly abelian groups as its output. The trade off is that the homotopy groups are very hard to compute. Such trade off has led to the blossoming of Algebraic Topology in its fullest. For instance, stable homotopy theory stems from a crucial fact called the Freudenthal suspension theorem, which states that such a sequence

$$\pi_n(X) \rightarrow \pi_{n+1}(\Sigma X) \rightarrow \cdots$$

eventually stabilizes for large enough n .

In this chapter we will closely study the n th homotopy groups such as its properties and develop tools to compute them.

4.1 The n th Homotopy Groups

We begin not with the definition of the homotopy groups, but rather a slight generalization of pointed spaces and maps between them.

Definition 4.1.1 (Pairs of Space) Let X be a topological space. A pair of space is a pair (X, A) where $A \subseteq X$ is a subspace of X . A map of pairs $f : (X, A) \rightarrow (Y, B)$ is a continuous map $f : X \rightarrow Y$ such that $f(A) \subseteq B$.

Definition 4.1.2 (Homotopy between Maps of Pairs) Let $f, g : (X, A) \rightarrow (Y, B)$ be maps of pairs. A homotopy between f and g is a homotopy $H : X \times [0, 1] \rightarrow Y$ such that $H(A \times [0, 1]) \subseteq B$.

Definition 4.1.3 (The n th Homotopy Groups) Let (X, x_0) be a pointed space. Define the n th homotopy group $\pi_n(X, x_0)$ to be

$$\pi_n(X, x_0) = \frac{\left\{ \gamma : (I^n, \partial I^n) \rightarrow (X, \{x_0\}) \mid \gamma \text{ is continuous} \right\}}{\simeq}$$

where we say that $f \simeq g$ if there exists a homotopy between f and g .

Notice that the definition coincides with that of the fundamental group when $n = 1$, and hence π_n is indeed a generalization.

Now we have defined the set $\pi_n(X, x_0)$ for a pointed space to have the word group in its name. We will also need to procure a canonical group structure on the set $\pi_n(X, x_0)$. This will be similar with that of the fundamental group.

Definition 4.1.4 (Concatenation) Let $n \geq 1$. Let (X, x_0) be a pointed space. Let $f, g : (I^n, \partial I^n) \rightarrow (X, x_0)$ be maps. Define the composition of f and g by the formula

$$(f \cdot g)(t_1, \dots, t_n) = \begin{cases} f(2t_1, t_2, \dots, t_n) & \text{if } 0 \leq t_1 \leq \frac{1}{2} \\ g(2t_1 - 1, t_2, \dots, t_n) & \text{if } \frac{1}{2} \leq t_1 \leq 1 \end{cases}$$

for $f, g \in \pi_n(X, x_0)$.

Notice that concatenation is really just the same concatenation between elements of the fundamental group but instead with more coordinates. The group structure on $\pi_n(X, x_0)$ uses concatenation and such a proof also uses the same homotopies as in Algebraic Topology 1, but with more coordinates.

Theorem 4.1.5 Let (X, x_0) be a pointed space and $n \geq 1$. The operation $\cdot$ on the equivalence classes in $\pi_n(X, x_0)$ is well defined and endows it with the structure of a group.

Proof We first show that the operation is well defined on $\pi_n(X, x_0)$. Suppose that $f_1 \stackrel{\partial}{\simeq} g_1 : (I^n, \partial I^n) \rightarrow (X, x_0)$ via the homotopy H_1 and $f_2 \stackrel{\partial}{\simeq} g_2 : (I^n, \partial I^n) \rightarrow (X, x_0)$ via the homotopy H_2 . Consider the map $H : I^n \times [0, 1] \rightarrow X$ defined by

$$H(x_1, \dots, x_n, t) = \begin{cases} H_1(2x_1, \dots, x_n, t) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \\ H_2(2x_1 - 1, \dots, x_n, t) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \end{cases}$$

Now when $t = 0$, we have that $H(x_1, \dots, x_n, 0) = f_1 \cdot f_2$. When $t = 1$, we have that $H(x_1, \dots, x_n, 1) = g_1 \cdot g_2$. Now notice that by definition of H_1 and H_2 , if one of $x_1, \dots, x_n$ is equal to 0 or 1, then H_1 and H_2 is constant and maps to x_0 . This means that H also has such property and hence H is a homotopy $(I, \partial I^n)$ to (X, x_0) .

We now have an appropriate binary operation on $\pi_n(X, x_0)$. It is clearly associative since the composition of maps are associativity and one can re-parametrize homotopies with different traversal speeds. I claim that the constant map $e_{x_0} : (I, \partial I^n) \rightarrow (X, x_0)$ defined by $e_{x_0}(x) = x_0$ is the identity. Let $f : (I^n, \partial I^n) \rightarrow (X, x_0)$ be arbitrary. Define the homotopy from $e_{x_0} \cdot f$ to f by

$$H(x_1, \dots, x_n, t) = \begin{cases} e_{x_0}(x_1, \dots, x_n) = x_0 & \text{if } 0 \leq x_1 \leq \frac{1-t}{2} \\ f\left(\frac{2s+t-1}{t+1}\right) & \text{if } \frac{1-t}{2} \leq x_1 \leq 1 \end{cases}$$

A similar homotopy proves that $f \cdot e_{x_0} \simeq f$. For the inverse, I claim that $\bar{f} : (I^n, \partial I^n) \rightarrow (X, x_0)$ defined by $\bar{f}(1 - x_1, \dots, x_n)$ is the inverse of f . Indeed, define a homotopy from $f \cdot \bar{f}$ to e_{x_0} by

$$H(x_1, \dots, x_n, t) = \begin{cases} e_{x_0}(x_1, \dots, x_n) = x_0 & \text{if } 0 \leq x_1 \leq \frac{t}{2} \text{ or } \frac{1-t}{2} \leq x_1 \leq 1 \\ f(2x_1 - t, x_2, \dots, x_n) & \text{if } \frac{t}{2} \leq x_1 \leq \frac{1}{2} \\ \bar{f}(2s + t - 1) & \text{if } \frac{1}{2} \leq x_1 \leq \frac{1-t}{2} \end{cases}$$

However, what makes each $\pi_n(X, x_0)$ for $n \geq 2$ different from the fundamental group $\pi_1(X, x_0)$ is the abelian group structure on $\pi_n(X, x_0)$.

Theorem 4.1.6 Let (X, x_0) be a pointed space. Then the n th homotopy group

$$\pi_n(X, x_0)$$

together with concatenation is abelian.

Proof Define a new operation $\star : \pi_n(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ by

$$[f] \star [g] = \begin{cases} f(t_1, 2t_2, \dots, t_n) & \text{if } 0 \leq t_1 \leq \frac{1}{2} \\ g(t_1, 2t_2 - 1, \dots, t_n) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

Such an operation clearly also defines an abelian group structure on $\pi_n(X, x_0)$ using the same argument. Now I want to prove that

$$([f] \star [g]) \star ([h] \star [k]) = ([f] \star [h]) \star ([g] \star [k])$$

This is true because

$$([f] \star [g]) \star ([h] \star [k]) = \begin{cases} f(2x_1, 2x_2, x_3, \dots, x_n) & \text{if } 0 \leq x_1, x_2 \leq \frac{1}{2} \\ g(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \text{ and } \frac{1}{2} \leq x_2 \leq 1 \\ h(2x_1 - 1, 2x_2, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq \frac{1}{2} \\ k(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1, x_2 \leq 1 \end{cases}$$

and

$$([f] \star [h]) \star ([g] \star [k]) = \begin{cases} f(2x_1, 2x_2, x_3, \dots, x_n) & \text{if } 0 \leq x_1, x_2 \leq \frac{1}{2} \\ h(2x_1 - 1, 2x_2, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq \frac{1}{2} \\ g(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \text{ and } \frac{1}{2} \leq x_2 \leq 1 \\ k(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1, x_2 \leq 1 \end{cases}$$

which are entirely the same. Now I claim that $\star = \star$. It is clear that both binary operations have the same identity element e_{x_0} . Now we have that

$$f \star g = (f \star 1) \star (1 \star g) = (f \star 1) \star (1 \star g) = f \star g$$

Finally, I claim that $\star$ is commutative. We have that

$$f \star g = (1 \star f) \star (g \star 1) = (1 \star g) \star (f \star 1) = g \star f = g \star f$$

Thus we conclude. ■

The above technique is actually called the Eckmann-Hilton argument. In particular, it shows that concatenation of paths need not be defined via the first coordinate. Any choice of coordinate to perform concatenation will result in the same group structure.

Example 4.1.7

The homotopy groups of the sphere is given by

$$\pi_n(S^1) \cong \begin{cases} \mathbb{Z} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof

We have seen that $\pi_0(S^1) = 0$ and $\pi_1(S^1) \cong \mathbb{Z}$.

When $n \geq 2$, suppose that $[f] \in \pi_n(S^1)$. Then we have a map $f : S^k \rightarrow S^1$. By the lifting criterion in Algebraic Topology 1, we may lift the map to the covering space $p : \mathbb{R} \rightarrow S^1$ since $0 = f_*(\pi_1(S^k)) \leq p_*(\pi_1(\mathbb{R})) = 0$. Hence f factors through a contractible space, and so is null homotopic. Hence $[f] = 0$. ■

Lemma 4.1.8

For any $n \in \mathbb{N}^+$, the two spaces $(I^n, \partial I^n)$ and (S^n, s_0) are homotopy equivalent.

Corollary 4.1.9

Let (X, x_0) be a pointed space. Then we have

$$\pi_n(X, x_0) = [(S^n, 1), (X, x_0)]_*$$

Therefore an alternate viewpoint of the homotopy groups is instead the collection of maps from the pointed n -sphere to the space X quotient homotopy. Indeed an n -dimensional sphere has an n -dimensional hole enclosed by the sphere itself. Therefore in order to detect n -dimensional holes in a space, we are permitted to try and fit n -spheres into the space.

Spheres are also advantageous for the definition of π_n because spheres only has an n -dimensional hole and no other holes in any dimension. Therefore we are capturing the minimal amount of information on n -dimensional holes without producing excess data.

The group structure of π_n is also easy to understand in terms of spheres.

Lemma 4.1.10

Let (X, x_0) be a pointed space. Let $f, g : S^n \rightarrow X$ be pointed maps. Then $[f] \cdot [g] \in \pi_n(X, x_0)$ is represented by the map

$$S^n \xrightarrow{\text{proj.}} S^n \vee S^n \xrightarrow{f \vee g} X$$

where the projection map is given by $S^n \vee S^n \cong \frac{S^n}{S^{n-1}}$ where $S^{n-1} \hookrightarrow S^n$ is the inclusion along the equator.

4.2 Functorial Properties of Homotopy

The homotopy groups also satisfy functorial properties similar to the fundamental group and the (co)homology groups.

Theorem 4.2.1 (Functoriality) Let (X, x_0) and (Y, y_0) be pointed spaces and let $f : (X, x_0) \rightarrow (Y, y_0)$ be a pointed map. Then the induced map

$$\pi_n(f) : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

defined by $[\gamma] \mapsto [f \circ \gamma]$ is a group homomorphism. Moreover, it satisfies the following functorial properties.

- If $g : (Y, y_0) \rightarrow (Z, z_0)$ is a pointed map then

$$\pi_n(g \circ f) = \pi_n(g) \circ \pi_n(f)$$

- If $\text{id}_{(X, x_0)} : (X, x_0) \rightarrow (X, x_0)$ is the identity map then

$$\pi_n(\text{id}_{(X, x_0)}) = \text{id}_{\pi_n(X, x_0)}$$

Proof Firstly, let us show that it is a group homomorphism. Let $\gamma_1, \gamma_2 \in \pi_n(X, x_0)$. We have that

$$\pi_n(f)([\gamma_1] \cdot [\gamma_2]) = [f \circ (\gamma_1 \cdot \gamma_2)] = [f \circ \gamma_1 \cdot f \circ \gamma_2] = \pi_n(f)([\gamma_1]) \cdot \pi_n(f)([\gamma_2])$$

where the second equality is true because homotopies are preserved under function composition. It remains to show associativity and unitality.

- Associativity: We have that

$$\pi_n(g \circ f)([\gamma]) = [g \circ f \circ \gamma] = \pi_n(g)([f \circ \gamma]) = (\pi_n(g) \circ \pi_n(f))([\gamma])$$

- Unitality: We have that

$$\pi_n(\text{id}_{(X, x_0)})([\gamma]) = [\text{id}_{(X, x_0)} \circ \gamma] = [\gamma] = \text{id}_{\pi_n(X, x_0)}([\gamma])$$

And so we conclude. ■

Similar to all other functorial properties we have seen throughout algebraic topology, a homeomorphism of spaces give an isomorphism on homotopy groups. Now that we know about category theory, we see that such a result does not depend on the definition of the homotopy groups or the (co)homology groups, but is in fact due to the functorial properties of each invariant.

Similar to (co)homology and the fundamental group, the homotopy groups are defined via a quotient with homotopy. Therefore we expect the homotopy groups to not be able to distinguish between homotopy equivalent spaces but not homeomorphic spaces.

Theorem 4.2.2 (Homotopy Equivalence) Let $(X, x_0), (Y, y_0)$ be pointed spaces and $f, g : (X, x_0) \rightarrow (Y, y_0)$ be pointed maps. If f and g are homotopic, then the induced maps

$$\pi_n(f) = \pi_n(g) : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

are equal. Moreover, if f is a homotopy equivalence, then $\pi_n(f)$ is an isomorphism.

Proof Let $[\gamma] \in \pi_n(X, x_0)$. Suppose that f and g are homotopic via $F : X \times I \rightarrow Y$. now define

$$H(x_1, \dots, x_n, t) = F(\gamma(x_1, \dots, x_n), t)$$

Then it is clear that $H(x_1, \dots, x_n, 0) = f \circ \gamma$ and $H(x_1, \dots, x_n, 1) = g \circ \gamma$. Thus $[f \circ \gamma] = [g \circ \gamma]$ and so we conclude that $\pi_n(f)([\gamma]) = \pi_n(g)([\gamma])$.

If f is a homotopy equivalence, then there exists $g : (Y, y_0) \rightarrow (X, x_0)$ such that $g \circ f \simeq \text{id}_{(X, x_0)}$ and $f \circ g \simeq \text{id}_{(Y, y_0)}$. By functoriality and homotopy equivalence, we have that

$$\pi_n(g) \circ \pi_n(f) = \text{id}_{\pi_n(X, x_0)} \quad \text{and} \quad \pi_n(f) \circ \pi_n(g) = \text{id}_{\pi_n(Y, y_0)}$$

and so we conclude. ■

Similar to the fundamental group, changing the base point via a path induces isomorphisms on homotopy groups with the same space but different base point.

Theorem 4.2.3 Let (X, x_0) and (X, x_1) be pointed spaces with the same base space. Let $u : I \rightarrow X$ be a path from x_0 to x_1 . Define the induced map

$$u_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

as follows. For $[\gamma] \in \pi_n(X, x_0)$ define $u_{\#}([\gamma])$ by first shrinking the domain of γ to a smaller concentric cube in I^n . Then inserting the path γ on each radical segment of the shell between the smaller cube and ∂I^n .

The construction of $u_{\#}$ is a group isomorphism. Moreover, it satisfies the following universal properties.

- If $v : I \rightarrow X$ is a path from x_1 to x_2 and $u \cdot v$ is the concatenation of these paths, then

$$(u \cdot v)_{\#} = u_{\#} \circ v_{\#}$$

- If c_{x_0} is the constant path from x_0 to x_0 then $(c_{x_0})_{\#}$ is the identity

Proposition 4.2.4 Let X_i for $i \in I$ be a family of path connected spaces. Then there are isomorphisms

$$\pi_n \left(\prod_{i \in I} X_i \right) \cong \prod_{i \in I} \pi_n(X_i)$$

Proposition 4.2.5 Let (X, x_0) and (X, x_1) be pointed spaces with the same base space. Let $u, v : I \rightarrow X$ be paths from x_0 to x_1 . If u and v are homotopic relative to end points then the induced

maps

$$u_{\#} = v_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

are equal.

This shows that if X is path connected, then $\pi_n(X, x_0)$ no longer depends on the choice of base point. Although there are no canonical isomorphisms between $\pi_n(X, x_0)$ and $\pi_n(X, x_1)$, we still forget about the base point in this case and write the homotopy groups as $\pi_n(X)$.

4.3 The Action of the Fundamental Group

Proposition 4.3.1

Let (X, x_0) be a pointed space. Let $n \geq 1$. Then the map $\pi_1(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ defined by

$$([u], [f]) \mapsto u_{\#}(f)$$

is a group action on $\pi_n(X, x_0)$.

When $n = 1$, the above group action is just left multiplication in $\pi_1(X, x_0)$.

4.4 Relative Homotopy Groups

Definition 4.4.1 (Triplets of Spaces) Let X be a topological space. A pointed pair of space is a triple (X, A_1, A_2) where $A_2 \subseteq A_1 \subseteq X$ are subspaces of X . A map between triplets of spaces $f : (X, A_1, A_2) \rightarrow (Y, B_1, B_2)$ is a map $f : X \rightarrow Y$ such that $f(A_1) \subseteq B_1$ and $f(A_2) \subseteq B_2$. If $A_2 = \{x_0\}$ is a single point we say that (X, A, x_0) is a pointed pair of spaces.

Definition 4.4.2 (Homotopy between Maps of Triplets) Let $f, g : (X, A_1, A_2) \rightarrow (Y, B_1, B_2)$ be maps triplets of spaces. A homotopy between f and g is a homotopy between $f : X \rightarrow Y$ and $g : X \rightarrow Y$, namely $H : X \times [0, 1] \rightarrow Y$ such that $H(A_1 \times [0, 1]) \subseteq B_1$ and $H(A_2 \times [0, 1]) \subseteq B_2$.

Definition 4.4.3 (The n th Relative Homotopy Groups) Let (X, A, x_0) be a pointed pair of space. Let $n \geq 2$. Regard I^{n-1} sitting inside I^n by $I^{n-1} = \{(x_1, \dots, x_n) \in I^n \mid x_n = 0\}$ and let $J^{n-1} = \partial I^n \setminus I^{n-1}$. Define the relative homotopy groups of the triple by

$$\pi_n(X, A, x_0) = \frac{\left\{ \gamma : (I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0) \mid \gamma \text{ is continuous} \right\}}{\simeq}$$

where we say that $f \simeq g$ if there exists a homotopy between f and g .

It is easy to see that $\pi_n(X, x_0, x_0) = \pi_n(X, x_0)$ so that homotopy groups are a special case of the relative homotopy groups.

Lemma 4.4.4 For any $n \in \mathbb{N}$, the two triplets $(I^n, \partial I^n, J^{n-1})$ and (D^n, S^{n-1}, s_0) are homotopy equivalent.

Theorem 4.4.5 Let (X, A, x_0) be a pointed pair of space. The composition law on definition 1.1.4 defines a group structure on $\pi_n(X, A, x_0)$ for $n \geq 2$. Moreover, $\pi_n(X, A, x_0)$ is abelian for $n \geq 3$.

4.5 Induced Maps of Relative Homotopy Groups

Theorem 4.5.1 Let (X, A, x_0) and (Y, B, y_0) be pointed pairs of spaces and $f : (X, A, x_0) \rightarrow (Y, B, y_0)$ a map. Then f induces a map on the relative homotopy groups

$$f_* : \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$$

for $n \geq 2$ satisfying the following functorial properties:

- f_* is a group homomorphism
- If $g : (Y, B, y_0) \rightarrow (Z, C, z_0)$ is a map, then

$$(g \circ f)_* = g_* \circ f_*$$

- If $\text{id}_{(X, A, x_0)}$ is the identity map on (X, A, x_0) , then

$$(\text{id}_{(X, A, x_0)})_* = \text{id}_{\pi_n(X, A, x_0)}$$

Theorem 4.5.2 Let $(X, A, x_0), (Y, B, y_0)$ be pointed pairs of spaces and $f, g : (X, A, x_0) \rightarrow (Y, B, y_0)$ be pointed maps. If f and g are homotopic, then the induced maps

$$f_* = g_* : \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$$

are equal. Moreover, if f is a homotopy equivalence, then f_* is an isomorphism.

TBA: change of base point isomorphisms.

4.6 Long Exact Sequence in Homotopy Groups

Lemma 4.6.1 (Compression Criterion) Let (X, A, x_0) be a pair of spaces with basepoint. Let $f : (D^n, S^{n-1}, *) \rightarrow (X, A, x_0)$ be a map. Then $[f] = [e_{x_0}] \in \pi_n(X, A, x_0)$ if and only if

$$(f : D^n \rightarrow X) \stackrel{S^{n-1}}{\simeq} (g : D^n \rightarrow X)$$

where g is any map such that $g(X) \subseteq A$.

Proof Suppose that the second criterion is satisfied. Then it clearly shows that $[f] = [g] \in \pi_n(X, A, x_0)$. Let $r : D^n \times I \rightarrow D^n$ be a deformation retract from D^n to $*$ in $S^{n-1} \subset D^n$. Consider the map $g \circ r : D^n \times I \rightarrow X$. When $t = 0$, this is the map g . When $t = 1$, $g \circ r$ factors through $*$ and so becomes a map $*$ $\rightarrow X$. In other words, it is the constant map e_{x_0} . Moreover, it $g \circ r$ has image in A and so in particular it sends S^{n-1} to A . Thus $g \circ r$ is a homotopy between e_{x_0} and g . We conclude that $[f] = [g] = [e_{x_0}]$.

Now suppose that $[f] = [e_{x_0}] \in \pi_n(X, A, x_0)$ is given by the homotopy $H : D^n \times I \rightarrow X$. This means that $H(D^n \times \{1\}) \subseteq \{x_0\} \subset A$ and $H(S^{n-1} \times I) \subset A$. Now $D^n \times I$ deformation retracts to the cup $D^n \times \{1\} \cup S^{n-1} \times I$ by radial projection from the center point of $D^n \times \{0\}$. Thus H can be converted into a map from $D^n \times \{1\} \cup S^{n-1} \times I$ to X . Then H is now a homotopy from f to a map $H(-, 1) : D^n \rightarrow X$ which has image in A , relative to S^{n-1} . Thus we conclude. ■

Theorem 4.6.2 Let X be a space and A, B be subspaces of X such that $B \subseteq A \subseteq X$. Let $x_0 \in B$. Then there is a long exact sequence in relative homotopy groups:

$$\cdots \longrightarrow \pi_n(A, B, x_0) \xrightarrow{i_*} \pi_n(X, B, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial_n} \pi_{n-1}(A, B, x_0) \longrightarrow \cdots \longrightarrow \pi_1(X, A, x_0)$$

where $i : (A, B, x_0) \rightarrow (X, B, x_0)$ and $j : (X, B, x_0) \rightarrow (X, A, x_0)$ are the inclusions and $\partial : \pi_n(X, A, x_0) \rightarrow \pi_{n-1}(A, B, x_0)$ is given by $[\gamma] \mapsto [\gamma|_{I^{n-1}}]$

Proof ■

TBA: Naturality of the sequence.

Theorem 4.6.3 Let (X, A, x_0) be a pointed pair of spaces. The relative homotopy groups and (absolute) homotopy groups of (X, A, x_0) fit into a long exact sequence

$$\cdots \longrightarrow \pi_{n+1}(X, A, x_0) \xrightarrow{\partial_{n+1}} \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial_n} \pi_{n-1}(A, x_0) \longrightarrow \cdots \longrightarrow \pi_0(X, x_0) \longrightarrow 0$$

where ∂_n is defined by $[f] \mapsto [f|_{I^{n-1}}]$ and i_* and j_* are induced by inclusions.

Note that even though at the end of the sequence group structures are not defined, exactness still makes sense: kernels in this case consists of elements that map to the homotopy class of the constant map.

4.7 Homotopical Connectivity

Definition 4.7.1 (n-Connected Space)

Let X be a space. We say that it is n -connected if

$$\pi_k(X, x_0) = 0$$

for $0 \leq k \leq n$ and some $x_0 \in X$.

Note that $\pi_0(X, x_0)$ implies that X is path connected. Hence the notion of n -connectedness does not depend on the base point by the change of base point isomorphism. In particular, $\pi_k(X, x_0) = 0$ for $0 \leq k \leq n$ and some $x_0 \in X$ if and only if $\pi_k(X, x_0) = 0$ for $0 \leq k \leq n$ for all $x_0 \in X$. (Hatcher)

Definition 4.7.2 (n-Connected Pair of Spaces)

Let (X, A) be a pair of space. We say that it is n -connected if the following are true.

- $\pi_k(X, A, x_0) = 0$ for $0 < k \leq n$ and all $x_0 \in A$.
- $\pi_0(\iota) : \pi_0(A) \rightarrow \pi_0(X)$ is surjective.

TBA: conditions in P.346 of Hatcher

Definition 4.7.3 (Weakly Contractible)

Let X be a space. We say that X is weakly contractible if

$$\pi_n(X) = 0$$

for all $n \geq 0$.

Definition 4.7.4 (Weak Homotopy Equivalence) We say that a map $f : X \rightarrow Y$ is a weak homotopy equivalence if it induces isomorphisms on all homotopy groups π_n on any choice of base point.

TBA: compression lemma in Hatcher

Proposition 4.7.5

Let X, Y be spaces and let $f : X \rightarrow Y$ be a weak homotopy equivalence. Then f induces bijections

$$[Z, X] \cong [Z, Y] \quad \text{and} \quad [Z, X]_* \cong [Z, Y]_*$$

for all CW-complexes Z .

5 CW Complexes

CW complexes are an important class of topological spaces. In fact, they envelop a larger number of topological spaces, and all the commonly seen topological spaces are usually CW complexes. CW complexes allow combinatorial methods to be applied so that calculations of the fundamental group (or even homology in the sequel) becomes possible.

5.1 CW Complexes as a Topological Space

The definition of CW complexes may seem daunting. But intuitively, it is built up by inductively attaching n -dimensional disc to the $(n-1)$ -skeleton by gluing the boundary of the discs to the skeleton.

Definition 5.1.1 (CW Complexes)

A CW complex is a space X defined inductively in the following way.

- A discrete set X^0 that is the zero-skeleton.
- Given the $n-1$ skeleton X^{n-1} , a collection of closed n -dimensional disks $\{D_\alpha^n | \alpha \in I_n\}$ such that $D_\alpha^n \cong B_1(0) \subseteq \mathbb{R}^n$ with $S_\alpha^{n-1} = \partial D_\alpha^n$ and a collection of attaching maps $\varphi_\alpha : S_\alpha^{n-1} \rightarrow X^{n-1}$. Now define

$$X^n = \left(X^{n-1} \amalg \coprod_{\alpha \in I_n} D_\alpha^n \right) / \sim$$

where $\sim$ is the equivalence relation $x \sim \varphi_\alpha(x)$ for all $x \in S_\alpha^{n-1}$ (The unions in X are all disjoint)

- Define $X = \bigcup_n X^n$ equipped with the weak topology: a set $A \subset X$ is open if and only if $A \cap X^n$ is open in X^n for every n .

We say that $e_\alpha^n = D_\alpha^n \setminus S_\alpha^{n-1}$ the open cells of X and D_α^n are the closed cells with boundary S_α^{n-1} . If the inductive construction terminates at the n th skeleton, we say that X is of dimension n .

The weak topology is just the quotient topology when the inductive build up of the CW complex terminates after finite steps.

Definition 5.1.2 (Characteristic Maps) Let X be a CW-complex. Let D_α^n be a closed cell of X . Define the characteristic map of D_α^n to be the map

$$\Phi_\alpha^n : D_\alpha^n \rightarrow X$$

given by the following diagram

$$\begin{array}{ccc} & X^{n-1} \amalg (\coprod_\alpha D_\alpha^n) & \\ \iota \nearrow & \downarrow q & \\ D_\alpha^n & \xrightarrow{\quad} & X^n \xrightarrow{\quad} X \end{array}$$

where X^n arises by identifying $x \sim \varphi_\alpha^n(x)$ meaning q is the quotient map induced by the identification.

Each of the φ_α^n defines Φ_α^n because the attaching maps describes how the boundary of a cell is attached and the remaining interior is mapped arbitrarily and continuously. Notice that the restriction of Φ_α^n to e_α^n is a homeomorphism to its image, which we also denote by e_α^n . We can therefore characterize CW-complexes in terms of the disjoint union of all cells of the form e_α^n .

Example 5.1.3

The following are CW complexes.

- S^n for each $n \in \mathbb{N}$.
- $\mathbb{RP}^n$ for each $n \in \mathbb{N}^+$.
- $\mathbb{CP}^n$ for each $n \in \mathbb{N}^+$.

Proposition 5.1.4 Let X be a CW complex. Then the following are true regarding the topology of X .

- A subset $A \subseteq X$ is open in X if and only if $\Phi_\alpha^{-1}(A)$ is open in D_α^n for all characteristic maps $\Phi_\alpha : D_\alpha^n \rightarrow X$.
- A subset $B \subseteq X$ is closed in X if and only if $\Phi_\alpha^{-1}(B)$ is closed in D_α^n for all characteristic maps $\Phi_\alpha : D_\alpha^n \rightarrow X$.

Proof Suppose that A is open in X . Then $A \cap X^n$ is open in X^n for every n . Then it is clear that $\Phi_\alpha^{-1}(A)$ is open by continuity of Φ_α . The same is true for closed subsets. Conversely, suppose that $A \subseteq X$ is such that $\Phi_\alpha^{-1}(A)$ is open for all characteristic maps $\Phi_\alpha : D_\alpha^n \rightarrow X$. Suppose further by induction that $A \cap X^{n-1}$ is open in X^{n-1} . X^n is defined as quotient space so A is open in X^n if and only if $p^{-1}(A)$ is open in $X^{n-1} \amalg \coprod_{\alpha \in I_n} D_\alpha^n$ where p is the projection map. It is clear that before taking quotient, the space is a disjoint union. By definition of open sets of disjoint unions, $A \cap X^n$ is open if and only if $A \cap X^{n-1}$ and $\phi_\alpha^{-1}(A)$ is open in D_α^n . But we already have these information so we are done. ■

Definition 5.1.5 (Subcomplex)

Let X be a CW complex. A CW subcomplex of X is a CW complex A such that every n -cell of A is an n -cell of X for all $n \in \mathbb{N}$.

Proposition 5.1.6 A compact topological subspace of a CW complex X is contained in a finite subcomplex.

Proof First we show that a compact set C in a CW complex X can meet only finitely many cells of X . Suppose the contrary. Then there is an infinite sequence of points $x_i \in C$ all lying in distinct cells. Then set $S = \{x_1, x_2, \dots\}$. Assume that $S \cap X^{n-1}$ is closed in X by induction on n . Then for each cell e_α^n of X , $\varphi_\alpha^{-1}(S)$ is closed in ∂D_α^n , and $\Psi_\alpha^{-1}(S)$ consists of at most one more point in D_α^n , so $\Psi_\alpha^{-1}(S)$ is closed in ∂D_α^n . Therefore $S \cap X^n$ is closed in X^n for each n . By the topology of X , S is closed. The same argument shows that any subset of S is closed thus S has the discrete topology. Since C is compact and S is a closed subset of C , S is compact. Thus S must be finite, a contradiction.

We prove by induction that any cell of X is contained in a finite subcomplex. Any 0-dimensional cell is itself a finite subcomplex. Assume this is true for each cell of X of dimension $n - 1$. Let e_α^n be an n -cell of X . Then ∂D_α^n is compact and so $\varphi_\alpha(\partial D_\alpha^n)$ is compact in X . By the above paragraph, φ_α^n meets only a finite number of cells in X^{n-1} . By induction each of those cells are contained in a finite subcomplex. Thus their union is also a finite subcomplex. Attaching e_α^n to this finite subcomplex shows that e_α^n is contained in a finite subcomplex. This completes the induction.

Since we know that C meets finitely many cells of X , and these finitely many cells are each contained in a finite subcomplex, the finite union of these finite subcomplexes is again a finite subcomplex and so we conclude. ■

Proposition 5.1.7 Every CW complex is a normal space and hence Hausdorff.

Proposition 5.1.8 Every CW complex is locally contractible.

Proposition 5.1.9 If $A \subset X$ is a subcomplex of a CW complex X , then there exists an open set $U \subset X$ with $A \subset U$ and such that U deformation retracts to A .

Proposition 5.1.10 Let X be a path connected topological space. Let n be fixed, and let $\phi_\alpha^n : S_\alpha^{n-1} \rightarrow X$ be a collection of attaching maps. Let

$$Y = \frac{X \amalg (\coprod_\alpha D_\alpha^n)}{x \sim \phi_\alpha^n(x)}$$

Let $x_0 \in X$ be a point. Then the following are true.

- If $n = 2$ then $\pi_1(Y, x_0) \cong \pi_1(X, x_0)/N$ where $N = N(\gamma_\alpha \phi_\alpha^2 \bar{\gamma}_\alpha)$ is the normalizer generated by the elements $\gamma_\alpha \phi_\alpha^2 \bar{\gamma}_\alpha$ and $\gamma_\alpha : I \rightarrow X$ a path from x_0 to $\phi_\alpha^2(s_0)$ and s_0 a basepoint of S_α^1 .
- If $n > 2$ then $\pi_1(Y, x_0) \cong \pi_1(X, x_0)$

Intuitively, attaching 2-cells to the CW-complex allows homotopy to occur on the hole that the 2-cell had filled up, which is why we further quotient the fundamental group out by a normal subgroup of homotopy classes. When $n \geq 3$, the attachment of n -cells no longer alter the amount of 1-dimensional holes since the attachment must have boundary an $(n - 1)$ -cell. This also leads nicely to the following theorem.

Theorem 5.1.11 Let X be a path connected CW complex with $x_0 \in X^2$. The inclusion map $\iota : X^2 \rightarrow X$ induces an isomorphism of fundamental groups

$$\pi_1(X^2, x_0) \cong \pi_1(X, x_0)$$

This is why we cannot distinguish higher dimensional topological properties just by studying the fundamental group. Which is a set of 1-dimensional objects, namely loops.

Definition 5.1.12 (Product Complexes)

Let X, Y be CW complexes. Define the product complex of X and Y to be the CW complexes defined via the following data.

- The underlying set is given by the Cartesian product $X \times Y$ of sets.
- Suppose that the characteristic maps of X are given by $\{\Phi_\alpha \mid \alpha \in I_X\}$ and the characteristic maps of Y are given by $\{\Psi_\beta \mid \beta \in I_Y\}$. Then the characteristic maps of $X \times Y$ is given by $\Phi_\alpha \times \Psi_\beta$.

In general, there is no reason to believe that the product topology has anything to do with the weak topology, lest for the following.

Proposition 5.1.13

Let X, Y be CW complexes. Suppose that one of the following holds.

- One of X or Y is compact.
- Both X and Y have countable many cells.

Then the product complex $X \times Y$ is homeomorphic to the product of X and Y as topological spaces.

Definition 5.1.14 (Cellular Maps)

Let X, Y be CW complexes. Let $f : X \rightarrow Y$ be a map. We say that f is a cellular map if $f(X^n) \subseteq Y^n$ for all $n \in \mathbb{N}$.

5.2 Admitting CW-Complex Structures

It is natural to ask that given a space and data mapping n -cells into the space, when will the space admit a CW complex structure. The answer is given by the following.

Theorem 5.2.1 Let X be a Hausdorff space and let $\Phi_\alpha : D_\alpha^n \rightarrow X$ be a collection of maps. Then these maps form the characteristic map of a CW-complex structure on X if and only if the following are true.

- Each Φ_α is injective on to the interior $(D_\alpha^n)^\circ \cong e_\alpha^n$.
- Each interior e_α^n are disjoint from one another and $X = \coprod_{\alpha, n} e_\alpha^n$
- Each $\Phi_\alpha(\partial D_\alpha^n)$ is contained in the union of a finite number of cells of dimension less than n
- A subset X is closed if and only if it meets the closure of each cell of X in a closed set.

Finally, we can relate CW complexes to covering space.

Proposition 5.2.2 Let X be a finite CW complex and $p : \tilde{X} \rightarrow X$ be a covering space of X . Then $\tilde{X}$ is also a CW complex.

Proof Let $\Phi_\alpha : D_\alpha^n \rightarrow X$ be one of the characteristic maps. By the lifting property, there are precisely D lifts of the characteristic map. Label them $\Phi_{\alpha,i} : D_{\alpha,i}^n \rightarrow \tilde{X}$. Note that the fourth requirement above is void since X is a finite CW-complex and that there are finitely many cells.

Now since $\Phi_\alpha = p \circ \Phi_{\alpha,i}$ is injective on the interior, so is $\Phi_{\alpha,i}$. Let $U \subseteq (D_{\alpha,i}^n)^\circ$ be a non-empty open subset. Let y in the image with $x = p(y) \in X$. Then there exists a small open neighbourhood $x \in V \subseteq e_\alpha^n$ that is evenly covered. This means that $p^{-1}(V) = \coprod_{j=1}^d V_j$ for some disjoint V_j and $V_j \cong V$. It follows that $V_i \cap p^{-1}(\Phi_\alpha(U)) \subseteq \Phi_{\alpha,i}(U)$ is an open neighbourhood of y inside $e_{\alpha,i}^n$. If $y \in \tilde{X}$ is an arbitrary point with image $x = p(y)$, then there exists unique n and α such that $x \in e_\alpha^n$. By the unique lifting property, $y \in e_{\alpha,i}^n$ for exactly one i . y cannot belong to some $e_{\beta,j}^m$ with $(n, \alpha) \neq (m, \beta)$ either as otherwise $x \in e_\beta^m$. Thus the first and second condition is shown.

For the third condition, if $y = \Phi_{\alpha,i}(z)$ for some point $z \in \partial D_{\alpha,i}^n$ on the boundary, then $p(y) = x \in e_\beta^m$ for some $m < n$ and some β . Thus $e_{\beta,j}^m$ for some unique j . ■

5.3 CW Complex Structure of Projective Spaces

Let us begin with the projective space. Let $n \geq 1$. Recall that the projective space is defined to be

$$\mathbb{RP}^n = \frac{\mathbb{R}^{n+1} \setminus \{0\}}{\sim}$$

where we say that $x \sim y$ if $x = \lambda y$ for some $\lambda \in \mathbb{R}^\times$. This set is then equipped with the quotient topology. Now for each point $[x_0 : \dots : x_n]$ in the projective space, choose the two representatives such that $|(x_0, \dots, x_n)| = 1$ so that we have

$$\mathbb{RP}^n = \frac{S^n}{\sim}$$

where $\sim$ is the same equivalence relation restricted to the sphere.

Proposition 5.3.1 Let $n \geq 1$. Then the real projective space $\mathbb{RP}^n$ has a CW structure given inductively as follows.

- $\mathbb{RP}^1 \cong S^1$ has the structure of a CW complex identical to that of S^1
- Suppose that $\mathbb{RP}^{n-1}$ admits a CW structure. Attach an n -cell D^n to $\mathbb{RP}^{n-1}$ via the quotient map

$$\partial D^n \cong S^{n-1} \rightarrow \frac{S^{n-1}}{\sim} \cong \mathbb{RP}^{n-1}$$

Then the resulting space is $\mathbb{RP}^n$ and thus has a CW complex structure.

In particular, the CW structure of $\mathbb{RP}^n$ consists of the following.

- For each $0 \leq k \leq n$, $\mathbb{RP}^n$ has one k -cell D^k
- For each k -cell D^k , the attaching map is defined inductively to be the quotient map

$$\partial D^k \cong S^{k-1} \rightarrow \mathbb{RP}^{k-1} \cong \frac{S^{k-1}}{\sim}$$

Proof The case $n = 1$ is already given. We can proceed to the inductive case. This means that we want to show that the space given by attaching D^n to $\mathbb{RP}^{n-1}$ in the above way is homeomorphic to $\mathbb{RP}^n$. Suppose that the resulting space is called X . Write the quotient map as

$$q : S^{n-1} \rightarrow \frac{S^{n-1}}{\sim} \cong \mathbb{RP}^{n-1}$$

The definition of CW complexes imply that

$$X = \frac{\mathbb{RP}^{n-1} \amalg D^n}{\sim'}$$

where $x \sim' q(x)$ for all $x \in S^{n-1} = \partial D^n$. I claim that $X \cong \frac{D^n}{\sim}$ where $\sim$ is the equivalence relation of $\mathbb{R}P^n = S^n / \sim$ restricted to the northern hemisphere $\{(x_0, \dots, x_n) \in S^n \mid x_n \geq 0\} \cong D^n$. Explicitly, this relation is given by $x \sim y$ if $x, y \in \partial D^n$ and $x = -y$. To construct a map $D^n / \sim \rightarrow X$, we need the universal property of $D^n / \sim$ as a quotient:

$$\begin{array}{ccc} D^n & \xrightarrow{v} & \frac{D^n}{\sim} \\ & \searrow \iota & \downarrow \exists! \\ & \mathbb{R}P^{n-1} \amalg D^n & \downarrow p \\ & & X \end{array}$$

Let $[x] = [y] \in D^n / \sim$. We want to show that $p(\iota(x)) = p(\iota(y))$. Since ι is just the inclusion, it suffices to show that $p(x) = p(y)$. If x and y are not in the boundary of D^n , then from the equivalence relation in $D^n / \sim$ we deduce that $x = y$ and hence $p(x) = p(y)$. If at least one of x or y lies in ∂D^n , then from the equivalence relation in $D^n / \sim$ we deduce that both x and y lies in $\partial D^n = S^{n-1}$ and moreover $x = -y$. In other words, x and y are antipodal. Now the equivalence relation $\sim'$ in X implies that $x \sim' q(x) = q(y) \sim' y$ since q is the projection mapping antipodal points to the same equivalence class. We conclude that $p(\iota(x)) = p(\iota(y))$. By the universal property of quotients, there exists a unique map $X \rightarrow D^n / \sim$.

To construct a map $X \rightarrow D^n / \sim$, we need the universal property of X as a quotient:

$$\begin{array}{ccc} D^n & \xrightarrow{\iota} \mathbb{R}P^{n-1} \amalg D^n & \xrightarrow{p} X \\ & \searrow v & \downarrow \exists! \\ & & \frac{D^n}{\sim} \end{array}$$

Let $[x] = [y] \in X$. We want to show that $v(x) = v(y)$. There are some cases to consider. If x and y both does not lie in S^{n-1} , then the equivalence relation $\sim'$ is not invoked and thus we conclude that $x = y$. Then $v(x) = v(y)$ follows. If exactly one of x or y is in $S^{n-1} = \partial D^n$, then without loss of generality assume $x \in S^{n-1}$. Notice that each equivalence class of X consists exactly of 2 elements. If $x \in S^{n-1}$ then $[x] = \{x, q(x)\}$ and we conclude that $q(x) = y$. Because q is the antipodal map, we conclude that $x = y$ or $x = -y$. If $x = y$ we conclude that $v(x) = v(y)$. If $x = -y$, then $v(x) = v(y)$ because of the equivalence relation of $\sim$ in $D^n / \sim$ (because $x \in S^{n-1} = \partial D^n$). Thus there exists a unique map $X \rightarrow D^n / \sim$.

By the universal property of the quotient $D^n / \sim$, there exists a unique map k such that the following diagram commutes:

$$\begin{array}{ccc} D^n & \xrightarrow{v} & \frac{D^n}{\sim} \\ & \searrow v & \downarrow \exists! k \\ & & \frac{D^n}{\sim} \end{array}$$

Since the identity map satisfies the description, this unique map k must be $k = \text{id}$. We conclude that the composition of the unique map $D^n / \sim \rightarrow X \rightarrow D^n / \sim$ is the identity map. Thus $D^n / \sim$ and X are homeomorphic. Now $D^n / \sim$ is actually homeomorphic to $\mathbb{R}P^n = S^n / \sim$ by considering D^n as the upper hemisphere as S^n . We conclude that $X \cong \mathbb{R}P^n$. ■

For those who have knowledge in pushouts, we have essentially showed that the following diagram:

$$\begin{array}{ccc} S^{n-1} & \hookrightarrow & D^n \\ \downarrow & & \downarrow \\ \mathbb{R}P^{n-1} & \hookrightarrow & \mathbb{R}P^n \end{array}$$

is a pushout diagram.

Recall that the definition of the complex projective space is $\mathbb{C}P^n = \frac{\mathbb{C}^{n+1} \setminus \{0\}}{\sim}$ where we say that $x \sim \lambda x$ for $\lambda \in \mathbb{C}^\times$. Now $\mathbb{C}^{n+1}$ is homeomorphic to $\mathbb{R}^{2n+2}$, and inside it sits the sphere S^{2n+1} . By restricting the

equivalence relation on $\mathbb{CP}^n$ to S^{2n+1} , we obtain an isomorphism

$$\mathbb{CP}^n \cong \frac{S^{2n+1}}{\sim}$$

where $x \sim -x$ for $x \in S^{2n+1}$. It is now clear that there is a projection $S^{2n-1} \rightarrow \mathbb{CP}^{n-1}$. On the other hand, we want to make $\mathbb{CP}^n$ into a quotient of D^{2n} .

Proposition 5.3.2 Let $n \geq 1$. Then the complex projective space $\mathbb{CP}^n$ has a CW structure given inductively as follows.

- $\mathbb{CP}^1 \cong S^2$ has the structure of a CW complex identical to that of S^2
- Suppose that $\mathbb{CP}^{n-1}$ admits a CW structure. Attach an $2n$ -cell D^{2n} to $\mathbb{CP}^{n-1}$ via the quotient map

$$\partial D^{2n} \cong S^{2n-1} \rightarrow \frac{S^{2n-1}}{\sim} \cong \mathbb{CP}^{n-1}$$

Then the resulting space is $\mathbb{CP}^n$ and thus has a CW complex structure.

In particular, the CW structure of $\mathbb{CP}^n$ consists of the following.

- For each $0 \leq k \leq n$, $\mathbb{CP}^n$ has one $2k$ -cell D^{2k}
- For each $2k$ -cell D^{2k} , the attaching map is defined inductively to be the quotient map

$$\partial D^{2k} \cong S^{2k-1} \rightarrow \mathbb{CP}^{2k-1} \cong \frac{S^{2k-1}}{\sim}$$

Proof The case $n = 1$ is given. We can proceed to the inductive case. This means that we want to show that the space given by attaching D^{2n} to $\mathbb{CP}^{n-1}$ is homeomorphic to $\mathbb{CP}^n$. Suppose that the resulting space is X . Write the quotient map as

$$q : S^{2n-1} \rightarrow \frac{S^{2n-1}}{\sim} \cong \mathbb{CP}^{n-1}$$

The definition of CW complexes imply that

$$X = \frac{\mathbb{CP}^{n-1} \amalg D^{2n}}{\sim'}$$

where $x' \sim q(x)$ for all $x \in S^{2n-1} \cong \partial D^{2n}$.

We digress and look at how $\mathbb{CP}^n$ can be obtained as a quotient of D^{2n} . Consider the map $j : D^{2n} \rightarrow S^{2n+1}$ defined by

$$j(z_1, \dots, z_n) = \left(z_1, \dots, z_n, \sqrt{1 - \sum_{k=1}^{n+1} |z_k|^2} \right)$$

This map is injective because the $(n+1)$ th coordinate is determined by the first n coordinates. Thus j gives a homeomorphism to its image $j(D^{2n}) = T$. Now I claim that there is a homeomorphism

$$\frac{T}{\sim} \cong \frac{S^{2n+1}}{\sim} \cong \mathbb{CP}^n$$

where $\sim$ is given by restricting the equivalence relation in $\mathbb{CP}^n$ to the respective domains. We have the following diagram:

$$\begin{array}{ccc} T & \xrightarrow{w} & \frac{S^{2n+1}}{\sim} \\ \downarrow v & & \\ \frac{T}{\sim} & & \end{array}$$

We need to apply the universal property of quotients for both the quotients. Explicitly, the equivalence classes of $T/\sim$ is given as follows.

- If the last complex coordinate of $a \in T$ is non-zero, then $[a] = \{a\}$.

- If the last complex coordinate of $a = \left(a_1, \dots, a_n, \sqrt{1 - \sum_{k=1}^{n+1} |a_k|^2}\right) \in T$ is zero, then $a \in S^{2n+1}$ implies that the $\sum_{i=1}^n |a_i|^2 = 1$ so that $a \in \partial D^{2n} \cong \partial T$. In which case $[a] = \{a, -a\}$.

We first wish to apply the universal property of $T/\sim$. Let $a, b \in T$ such that $[a] = [b] \in S^{2n+1}/\sim$. We want to show that $v(a) = v(b)$. But $[a] = [b]$ implies $a = b$ or $a = -b$. If $a = b$ then we are done. If $a = -b$, then because the last coordinate of a and b are real and non-negative, we must conclude that the last coordinate of a and b are zero. In that case we have $v(a) = [a] = \{a, -a\} = \{a, b\} = [b] = v(b)$ and so we are done. By the universal property of quotients, there exists a unique map $T/\sim \rightarrow S^{2n+1}/\sim$. Now let $c, d \in T$ such that $[c] = [d] \in T/\sim$. We want to show that $w(c) = w(d)$. There are two cases. If the last complex coordinate of $c \in T$ is non-zero, then $\{c\} = [c] = [d] = \{d\}$ so that $c = d$. Then $w(c) = w(d)$ is true. Now if the last complex coordinate of $c \in T$ is zero, then $\{c, -c\} = [c] = [d] = \{d, -d\}$. Hence either $c = d$ or $c = -d$. Again if $c = d$ then we are done. If $c = -d$, then $w(c) = \{c, -c\} = \{c, d\} = [d] = w(d)$ and so we conclude. We conclude that $T/\sim$ and $S^{2n+1}/\sim$ are homeomorphic. Thus we have that

$$\mathbb{CP}^n \cong \frac{S^{2n+1}}{\sim} \cong \frac{D^{2n}}{\sim}$$

Back to the main proof. I claim that $X \cong D^{2n}/\sim$ whence we can conclude that $X \cong \mathbb{CP}^n$. To construct a map $D^{2n}/\sim \rightarrow X$, we need the universal property of $D^{2n}/\sim$ as a quotient:

$$\begin{array}{ccc} D^{2n} & \xrightarrow{\alpha} & D^{2n}/\sim \\ \downarrow \iota & & \downarrow \exists! \\ \mathbb{CP}^{n-1} \amalg D^{2n} & \xrightarrow{\beta} & X \end{array}$$

Let $[a] = [b] \in D^{2n}/\sim$. We want to show that $\beta(\iota(a)) = \beta(\iota(b))$. There are two cases. If the last complex coordinate is non-zero, then $a = b$ and so we are done. Otherwise, $a = -b$ and $a, b \in \partial D^{2n} \cong S^{2n-1}$. Then $\beta(-b) = [-b] = \{-b, b\} = \{a, b\} = [a] = \beta(a)$ and so we are done. By the universal property of quotients, we obtain a unique map $D^{2n}/\sim \rightarrow X$.

To construct a map $X \rightarrow D^{2n}/\sim$, we appeal to the universal property of X as a quotient. Let $[c] = [d] \in X$. We want to show that $\alpha(c) = \alpha(d)$. If $c \in \partial D^{2n} \cong S^{2n-1}$, then either $c = d$ or $c = -d$. In both cases, $\alpha(c) = \alpha(d)$ is clear. If $c \notin \partial D^{2n}$, then $c = d$ and so again we are done. We conclude that $D^{2n}/\sim \cong X$ are homeomorphic, and so we are done. $\blacksquare$

Again for those who have knowledge in pushouts, we have essentially showed that the following diagram:

$$\begin{array}{ccc} S^{2n-1} & \hookrightarrow & D^{2n} \\ \downarrow & & \downarrow \\ \mathbb{CP}^{n-1} & \hookrightarrow & \mathbb{CP}^n \end{array}$$

is a pushout diagram where $S^{2n-1} \rightarrow \mathbb{CP}^{n-1} \cong \frac{S^{2n-1}}{\sim}$ is the quotient map and $D^{2n} \rightarrow \mathbb{CP}^n$ is also the quotient map defined by the composition:

$$D^{2n} \hookrightarrow S^{2n+1} \twoheadrightarrow \mathbb{CP}^n$$

Using the CW complex structure of projective spaces, we can take the limit of n when it goes to infinity to obtain a new space.

Definition 5.3.3 (Real Projective Space) Define the real infinite projective space $\mathbb{RP}^\infty$ inductively as follows. For each $n \in \mathbb{N}$, the $(n-1)$ th skeleton is given by $\mathbb{RP}^{n-1}$, and one attaches an n -cell via the pushout

$$\begin{array}{ccc}
S^{n-1} & \hookrightarrow & D^n \\
\downarrow & & \downarrow \\
\mathbb{RP}^{n-1} & \hookrightarrow & \mathbb{RP}^n
\end{array}$$

to obtain $\mathbb{RP}^n$. This process is repeated for each $n \in \mathbb{N}$.

Definition 5.3.4 (Complex Projective Space) Define the complex infinite projective space $\mathbb{CP}^\infty$ inductively as follows. For each $n \in \mathbb{N}$, the $(2n-1)$ th skeleton is equal to the $(2n-2)$ th skeleton which is given by $\mathbb{CP}^{n-1}$, and one attaches a $2n$ -cell via the pushout

$$\begin{array}{ccc}
S^{2n-1} & \hookrightarrow & D^{2n} \\
\downarrow & & \downarrow \\
\mathbb{CP}^{n-1} & \hookrightarrow & \mathbb{CP}^n
\end{array}$$

to obtain $\mathbb{CP}^n$. This process is repeated for each $n \in \mathbb{N}$.

5.4 Computation of the Fundamental Group of CW Complexes

Recall that the fundamental group cannot distinguish higher dimensional topological properties, so without loss of assumption we can take all CW complexes in this section to be 2-skeletons. Without loss of generality, we can take the 2-skeleton to be path-connected.

The procedure to compute the fundamental group of this CW complex X is as follows.

1. Find a spanning tree T of X^1 . Let A be the set of all edges not in T . Pasting an edge $e \in A$ to T gives a subgraph that is homotopic to a circle S^1 . We can thus describe the fundamental group of X^1 as generated by these subgraphs: $\pi_1(X^1, x_0) \cong *_{e \in A} \mathbb{Z}$.
2. Let $e_\alpha^2 \subset X$ be a 2-cell and $\phi_\alpha : S_\alpha^1 \rightarrow X^1$ be the attaching map. Recall that $\gamma_\alpha(t) = \phi_\alpha(e^{2\pi it})$ is a loop, and hence homotopic to a loop consisting of edges in X^1 . Let $x_1 \in \phi_\alpha(S_\alpha^1)$ and let $g_\alpha : I \rightarrow X^1$ be a path with $g_\alpha(0) = x_0$ and $g_\alpha(1) = x_1$. Then

$$\omega_\alpha = [g_\alpha * \gamma_\alpha * \bar{g}_\alpha] \in \pi_1(X^1, x_0)$$

and therefore corresponds to a reduced word $u_\alpha \in A$. Then $\pi_1(X, x_0) \cong \pi_1(X^1, x_0) / \langle u_\alpha \rangle$.

Theorem 5.4.1 For every group G , there exists a path connected two dimensional CW complex X_G such that $\pi_1(X) \cong G$.

Proof Every group is the quotient of a free group, and so has a set of generators and relations. Write

$$G = \langle g_\alpha \mid r_\beta \rangle$$

for some indexing sets $\alpha \in I$ and $\beta \in J$. Construct X_G from $\bigvee_{\alpha \in I} S_\alpha^1$ by attaching 2-cells e_β^2 by the loops specified by the relations r_β . By 5.1.10, X_G has fundamental group precisely G . Thus we are done. ■

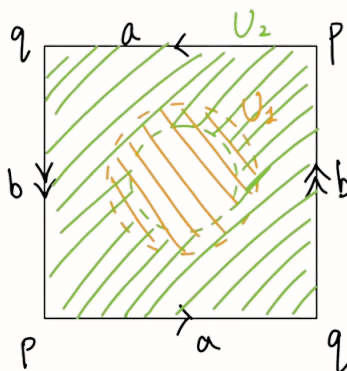
Let us see some examples of computing the fundamental group by exhibiting a CW-complex of the space. We have seen that the fundamental group of $\mathbb{RP}^2$ is $\mathbb{Z}/2\mathbb{Z}$ using the deck group. We can also combine Seifert-van Kampen and exhibit a CW-complex of $\mathbb{RP}^2$ to compute the fundamental group.

Theorem 5.4.2 The fundamental group of $\mathbb{RP}^2$ is

$$\pi_1(\mathbb{RP}^2, x) \cong \mathbb{Z}/2\mathbb{Z}$$

for any choice of base point $x \in \mathbb{Z}/2\mathbb{Z}$.

Proof Notice that $\mathbb{RP}^2$ is path connected so the choice of base point does not matter. Consider the following open cover of $\mathbb{RP}^2$.



Notice that the green part U_2 which is a Möbius band deformation retracts to a circle with paths $a \cdot b \cdot a \cdot b$ while the orange disc U_1 is contractible. They have intersection $U_1 \cap U_2$ that deformation retracts to S^1 . Choose base point $x_0 \in U_1 \cap U_2$. Using Seifert-van Kampen's theorem, we have that

$$\pi_1(\mathbb{RP}^2, x_0) \cong 1 *_{\pi_1(U_1 \cap U_2, x_0)} \mathbb{Z}$$

It remains to see how the amalgamated product is defined. Let $[\omega]$ be a generator of $\pi_1(U_1 \cap U_2, x_0)$. Then ω is a simple loop in $U_1 \cap U_2$. Notice that in U_1 , ω deformation retracts to the constant loop and in U_2 , ω deformation retracts to $a \cdot b \cdot a \cdot b$ the boundary. This means that $a \cdot b \cdot a \cdot b$ is identified as the identity by definition of the amalgamated product. Since $a \cdot b$ is a generator of $\pi_1(U_2, x_0)$, we have that

$$\pi_1(\mathbb{RP}^2, x_0) \cong \frac{\langle a \cdot b \rangle}{\langle a \cdot b \cdot a \cdot b \rangle} \cong \mathbb{Z}/2\mathbb{Z}$$

and so we conclude. ■

A similar method can be employed for the torus and the Klein bottle.

6 Whitehead's Theorem and CW Approximations

6.1 Whitehead's Theorem

Theorem 6.1.1 (Whitehead's Theorem) If X and Y are CW-complexes and $f : X \rightarrow Y$ is a weak homotopy equivalence, then f is a homotopy equivalence.

TBA: extension lemma in Hatcher.

Corollary 6.1.2 If X and Y are CW-complexes with $\pi_1(X) = \pi_1(Y) = 0$ and $f : X \rightarrow Y$ induces isomorphisms on homology groups H_n for all n , then f is a homotopy equivalence.

6.2 Cellular Approximations

Definition 6.2.1 (Cellular Approximations) Let X and Y be CW-complexes. We say that $f : X \rightarrow Y$ has a cellular approximations if f is homotopic to a cellular map $f' : X \rightarrow Y$.

To this end we need to revisit the notion of polyhedra.

Definition 6.2.2 (Convex Polyhedra) Let $n \in \mathbb{N}$. A convex polyhedra is a subset S of $\mathbb{R}^n$ of the form

$$S = \left\{ x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid \sum_{k=1}^n a_{k,1}x_k \leq b_1, \dots, \sum_{k=1}^n a_{k,s}x_k \leq b_s \right\}$$

for some $a_{1,1}, \dots, a_{n,s}, b_1, \dots, b_s \in \mathbb{R}$.

Lemma 6.2.3 Let X be a CW complex. Let $Y \subseteq X$ be a CW subcomplex. Let Z be obtained by attaching a cell e^k to Y . Let $f : I^n \rightarrow Z$ be a map. Then there exists a map $g : I^n \rightarrow X$ such that

$$f \stackrel{f^{-1}(Y)}{\simeq} g$$

and g is such that the following is true. There exists a simplex $\Delta^k \subset e^k$ and polyhedra $P_1, \dots, P_d$ such that g is the (possibly empty) union

$$g^{-1}(\Delta^k) = \bigcup_{k=1}^n$$

and $g|_{P_t}$ is the restriction of a linear surjection $\mathbb{R}^n \rightarrow \mathbb{R}^k$ for all t .

Theorem 6.2.4 (Cellular Approximation Theorem)

Let X, Y be CW complexes. Let $f : X \rightarrow Y$ be a map. Then the following are true.

- f is homotopic to a cellular map (called its cellular approximation).
- If f is homotopic to a cellular map f' and f is already cellular for some subcomplex $A \subseteq X$, then the homotopy to its cellular approximation can be taken such that $f'|_A = f|_A$.

Example 6.2.5

Let $0 \leq n < m$. Then $\pi_n(S^m) = 0$.

Proof

Let $[f] \in \pi_n(S^m)$. By the above theorem, we know that $f : S^n \rightarrow S^m$ is homotopic to a cellular map, where S^n is given the CW structure of a 0-cell and an n -cell. But the n -skeleton of S^m only consists of its 0-cell. Hence the cellular approximation of f is null-homotopic. Hence $[f] = 0$. ■

Example 6.2.6

$S^\infty = \text{colim}_{n \in \mathbb{N}} S^n$ is weakly contractible and contractible.

Proof

Let $n \in \mathbb{N}$. Let $[f] \in \pi_n(S^\infty)$. Then $f : S^n \rightarrow S^\infty$ has a compact image since S^n is compact. By definition of the colimit, we have that $\{S^n \mid n \in \mathbb{N}\}$ is an open cover of S^∞ , hence the image of f lies in $S^{n_1}, \dots, S^{n_k}$ for some $n_1, \dots, n_k \in \mathbb{N}$. Let $m = \max\{n_1, \dots, n_k, n + 1\}$. Since the colimit is a direct limit and each map is an inclusion, we have that f factors into the following:

$$\begin{array}{ccc} S^n & \xrightarrow{f} & S^\infty \\ & \searrow g & \nearrow \\ & S^m & \end{array}$$

But $m > n$ since $m \geq n + 1$, and we know that $\pi_n(S^m) = 0$ for $m > n$. Hence g is null homotopic, and so is f . Since S^∞ is a CW complex, by Whitehead's theorem, S^∞ is contractible. ■

Theorem 6.2.7 (Relative Cellular Approximation) Any map $f : (X, A) \rightarrow (Y, B)$ between pairs of CW-complexes has a cellular approximation.

Corollary 6.2.8 Let $A \subset X$ be CW-complexes and suppose that all cells $X \setminus A$ have dimension larger than n . Then (X, A) is n -connected.

Corollary 6.2.9 Let X be a CW complex and let X^n be its n -skeleton. Then (X, X^n) is n -connected. Moreover, the inclusion $X^n \hookrightarrow X$ induces an isomorphism

$$\pi_k(X^n) \rightarrow \pi_k(X)$$

for $0 \leq k < n$ and a surjection for $k = n$.

6.3 CW Approximations

Definition 6.3.1 (CW Approximation) Let X be a space. A CW approximation of X is a weak homotopy equivalence $f : Z \rightarrow X$ where Z is a CW complex.

The goal of this section is that every space has a CW approximation. The given homotopy equivalence makes this notion powerful because this means that for any space X , there exists a CW-complex such that X and Z are homotopy equivalent, and moreover, has isomorphic homotopy, homology and cohomology groups.

Definition 6.3.2 (CW Model) Let (X, A) be a non-empty pair of CW-complexes. An n -connected CW model of (X, A) is an n -connected CW pair (Z, A) together with a map $f : Z \rightarrow X$ with $f|_A = \text{id}_A$ such that

$$f_* : \pi_i(Z) \rightarrow \pi_i(X)$$

is an isomorphism for $i > n$ and an injection for $i = n$ for any choice of base point.

Theorem 6.3.3 For any non-empty pair (X, A) of CW-complexes, there exists an n -connected model (Z, A) . Moreover, Z can be built from A by attaching cells of dimension greater than n .

Theorem 6.3.4 Every pair of spaces (X, A) has a CW approximation. Such a CW approximation is unique up to homotopy equivalence.

6.4 Eilenberg-MacLane Spaces

Definition 6.4.1 (Eilenberg-MacLane Space)

Let G be a group and $n \in \mathbb{N}$. We say that a space X is an Eilenberg-MacLane space of type $K(G, n)$ if

$$\pi_k(X) = \begin{cases} K(G, n) & \text{if } k = n \\ 0 & \text{otherwise} \end{cases}$$

We often denote this space X directly by $X = K(G, n)$.

Proposition 6.4.2

Let G be a group. Then there exists a $K(G, 1)$ -CW complex.

Proposition 6.4.3

Let G be an abelian group and $n \geq 2$. Then there exists a $K(G, n)$ -CW complex. Moreover, it is unique up to homotopy equivalence.